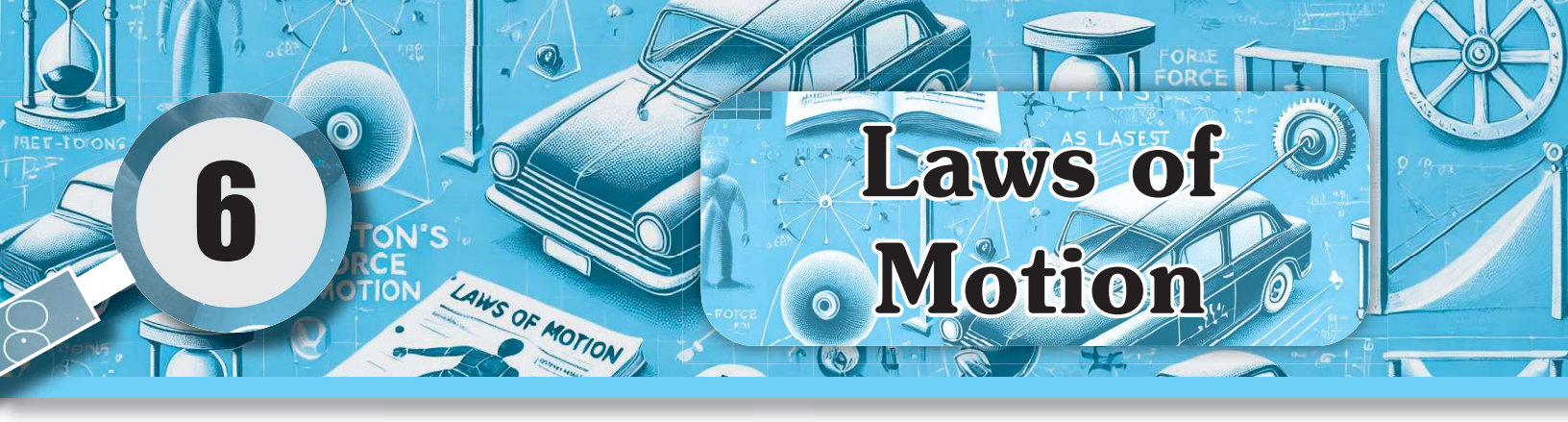


State of Body : Inertia

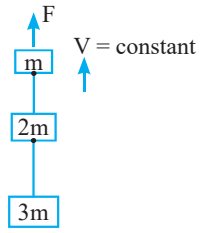
1. A boy sitting on the topmost berth in the compartment of a train which is just point to stop on a railway station, drops an apple aiming at the open hand of his brother sitting vertically below his hands at a distance of about 2 meter The apple will fall:
   (*a*) Precisely on the hand of his brother.
   (*b*) Slightly away from the hand of his brother in the direction of motion of the train.
   (*c*) Slightly away from the hand of his brother in the direction opposite to the direction of motion of the train.
   (*d*) None of the above
2. When an object is in equilibrium state then
   (*a*) It must be at rest
   (*b*) No force is acting on it
   (*c*) Its net acceleration must be zero
   (*d*) All of these
3. When an object is at rest
   (*a*) Force is required to keep it in rest state
   (*b*) No force is acting on it
   (*c*) A large number of forces may be acting on it which balance each other
   (*d*) It is in vacuum
4. A body released from a balloon rising up continues to move up along with the balloon due to inertial of motion.
   (*a*) True (*b*) False
   (*c*) Partially true (*d*) None of the above
5. Unit of inertia is same as unit of :-
   (*a*) Mass (*b*) Force
   (*c*) Di-electric constant (*d*) Angle

Newton's First Law of Motion or Law of Inertia

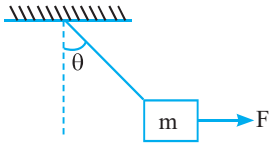
6. An athlete does not come to rest immediately after crossing the winning line due to the
   (*a*) Inertia of rest (*b*) Inertia of motion
   (*c*) Inertia of direction (*d*) None of these
7. Why does the horse rider falls forward when a horse at full gallop stops suddenly?
8. **Assertion (A):** If a body is moving with some fixed velocity, then there must be presence of Net force acting on Body.
   Reason (R): It is only the Net force which can change state of body from rest to motion
   (*a*) If both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of the Assertion (A)
   (*b*) If both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A)
   (*c*) If Assertion (A) is incorrect but Reason (R) is correct
   (*d*) If both Assertion (A) and Reason (R) are incorrect
9. **Assertion (A):** Inertia is proportional to mass of object [higher the mass, higher will be it's inertia]
   Reason (R): Inertia is physical quantity that doesn't have any unit
   (*a*) If both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of the Assertion (A)
   (*b*) If both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A)
   (*c*) If Assertion (A) is correct but Reason (R) is incorrect
   (*d*) If both Assertion (A) and Reason (R) are incorrect
10. Newton's first law is a special case of second law.
    (*a*) True (*b*) False
    (*c*) Partially true (*d*) None of the above

Tension in Equilibrium

11. Three blocks with masses m, $2m$, and $3m$ are connected as shown in the diagram. A force F is applied upward on the block with mass m, and the system moves with constant velocity V. Find the net force acting on the middle block of mass $2m$.



12. Find 'θ' angle by string from vertical and tension T in string where object of mans m is in equilibrium state.

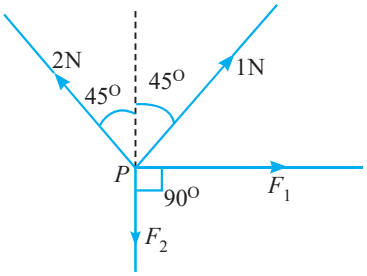


13. A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the mass, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is (Take, g = 10 ms^2)

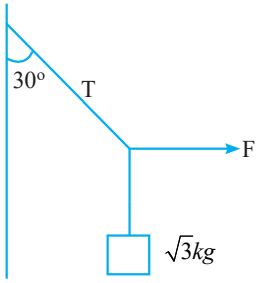
(*a*) 70 N (*b*) 200 N

(*c*) 100 N (*d*) 140 N

14. Four forces are acting at a point *P* in equilibrium as shown in figure. The ratio of force F_1 to F_2 is 1 : *x* where *x* =.....



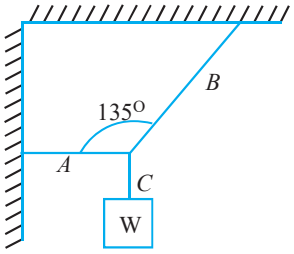
15. A block of $\sqrt{3}$ kg is attached to string whose other end is attached to the wall. An unknown force *F* is applied so that the string makes an angle of 30° with wall. The tension T is



(*a*) 15 N (*b*) 10 N

(*c*) 25 N (*d*) 20 N

16. A block of weight W is supported by three strings as shown in figure. Which of the following relations is true for tension in the strings? (Here T_1, T_2 and T_3 are the tension in the strings *A, B* and *C* respectively)



(*a*) $T_1 = T_2$ (*b*) $T_1 = T_3$

(*c*) $T_2 = T_3$ (*d*) $T_1 = T_2 = T_3$

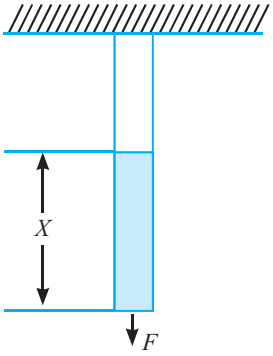
17. A weight Mg is suspended from the middle of a rope whose ends are at the same level. The rope is no longer horizontal. The minimum tension required to completely straighten the rope is

(*a*) Mg/2 (*b*) Mg cos 8

(*c*) 2 mg cos 8 (*d*) Infinitely large

Tension in a Heavy Rope

18. A vertical force *F* is applied at one end of a uniform rope of mass *M* and length *L*. Find out tension in the rope as a function of *x*



(*a*) $F + Mg$ (*b*) $F + \frac{MgL}{x}$

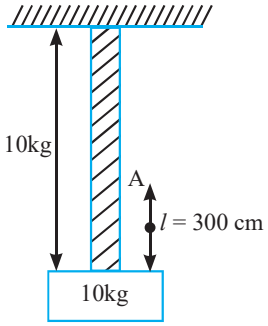
(*c*) $\frac{FL + Mgx}{L}$ (*d*) $\frac{Fx + MgL}{L}$

19. A uniform rope of mass *M* and length *L* is fixed at its upper end vertically from a rigid support. Then the tension in the rope at the distance *l* from the rigid support is

(*a*) $Mg\frac{L}{L+l}$ (*b*) $\frac{Mg}{L}(L-l)$

(*c*) Mg (*d*) $\frac{l}{L}Mg$

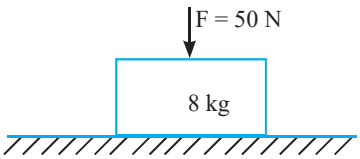
20. The adjoining figure shows a block of mass 10 kg connected to free end of a rope of mass 10 kg and length 10 m. The tension of the rope at point A is (g = 10m /s^2)



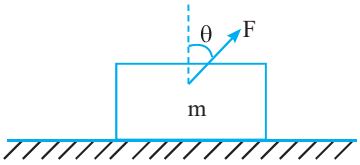
(*a*) 170 N (*b*) 30 N (*c*) 130 N (*d*) 70 N

Contact Force

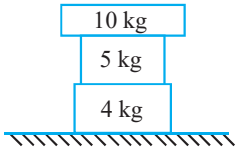
21. An 8 kg block is placed on the ground, and a vertical downward force of 50 N is applied to the block. Find the contact force (normal force) between the 8 kg block and the ground.



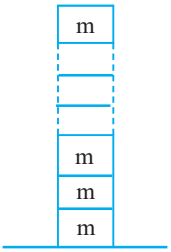
22. A block of mass *m* is placed on the ground. A force *F* is applied at an angle θ to the horizontal. Find the contact force (normal force) exerted by the block on the ground.



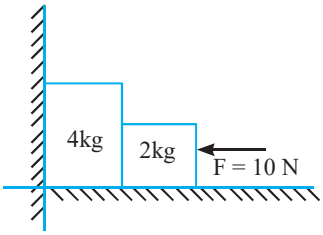
23. Three blocks of masses 4 kg, 5 kg, and 10 kg are stacked on top of each other, with the 4 kg block at the bottom, the 5 kg block in the middle, and the 10 kg block on top. Find the force applied by the 5 kg block on the 10 kg block.



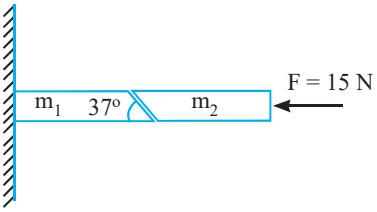
24. There are 50-identical block placed on over other shown in figure. Find contact force between 2nd and 3rd block.



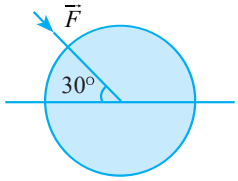
25. A 4 kg block is placed next to a 2 kg block, with a horizontal force of 10 N applied to the 2 kg block. The blocks are in contact with each other and are pushed against a wall as shown in the diagrame. Find the contact force between the 4 kg block and the 2 kg block..



26. Find contact force between object m_1 and m_2



27. As shown in figure, a 70 kg garden roller is pushed with force of $\vec{F} = 200$ N at an angle of 30° with horizontal. The normal reaction on the roller is



(*a*) 800 N (*b*) 600 N (*c*) $200\sqrt{3}N$ (*d*) $800\sqrt{2}N$

28. Ten one-rupee coins are put on top of each other on a table. Each coin has mass m.

Find the magnitude (is mgN) and direction of

(i) The force on the 7th coin (counted from the bottom due to all the coins on its top).

(ii) The force on the 7th coin by the eighth coin,

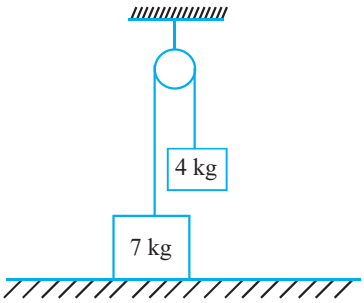
(iii) The reaction of the 6th coin on the 7th coin.

(*a*) 3, 3, –4 (*b*) –4, 3, 3

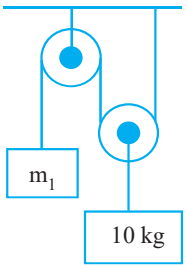
(*c*) 3, –4, 3 (*d*) None of these

Static Pulley Block System

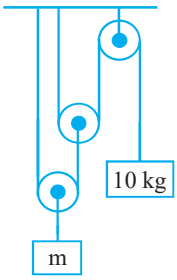
29. Find contact force between ground and 7 kg block as shown in figure.



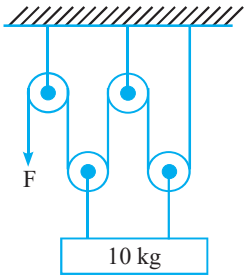
30. Find m_1 so that system is in equilibrium.



31. Find m = ?? if system is at rest?



32. F = ? So that system will be in equilibrium

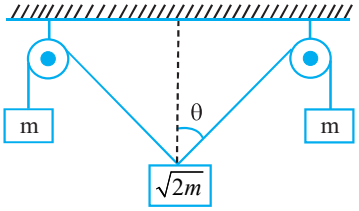


(*a*) 10 N (*b*) 25 N (*c*) 100 N (*d*) 400 N

33. Mechanical Advantage in above is

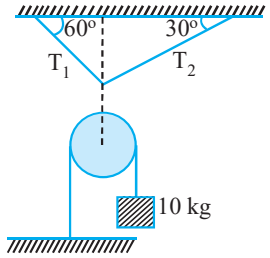
(*a*) 4 (*b*) 10 (*c*) $\frac{1}{4}$ (*d*) 1

34. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be



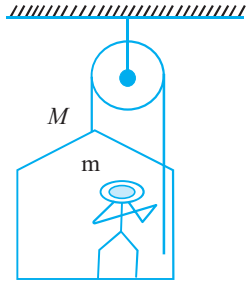
(*a*) 0° (*b*) 30° (*c*) 45° (*d*) 60°

35. In the arrangement as Shown, tension T_2 is (g = 10 m/s^2)



(*a*) 50 N (*b*) 100 N (*c*) $50\sqrt{3}N$ (*d*) $100\sqrt{3}$ N

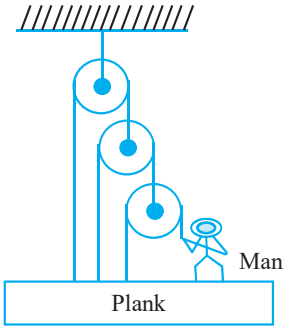
36. A man of mass *m* stands on a frame of mass *M*. He pulls a light rope, which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in uniform motion, what force must the man exert on the rope ?



(*a*) $\frac{(M+m)g}{2}$ (*b*) (M + m)g

(*c*) (M - m)g (*d*) (M + 2m)g

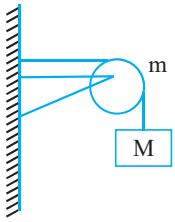
37. A man (mass *m*) hold himself and plank (mass *M*) in equilibrium with the help of 3 pulley + string system. The force exerted by man upon rope is



(*a*) $\frac{(M+m)g}{2}$ (*b*) $\frac{(M+m)g}{8}$

(*c*) $\frac{(M+m)g}{6}$ (*d*) $\frac{(M+m)g}{5}$

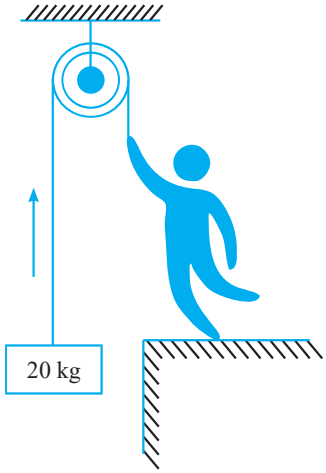
38. A string of negligible mass going over a clamped pulley of mass (m)supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by



(*a*) $\sqrt{2}$Mg (*b*) $\sqrt{2}$mg

(*c*) $g\sqrt{(M+m)^2+m^2}$ (*d*) $g\sqrt{(M+m)^2+M^2}$

39. Mass of a block is 20 kg. A man of mass 60 kg raises it with constant velocity as shown in the figure force exerted by man on the floor will be:-

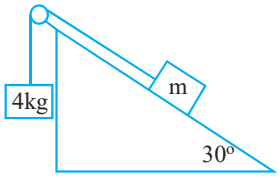


(*a*) 400 N (*b*) 600 N

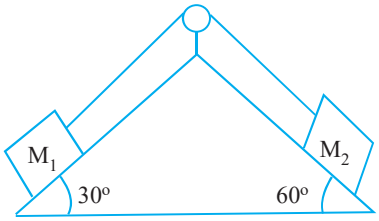
(*c*) 200 N (*d*) None of the above

Smooth Inclined Plane

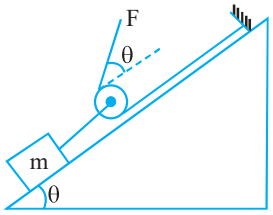
40. If system is in equilibrium then find value of m??



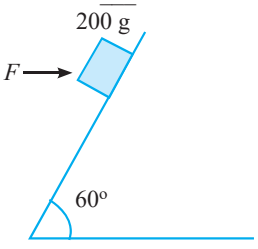
41. Find relation between m_1 and m_2 so that system is in equilibrium.



42. Find *F* so that object of mass *m* will be at rest on smooth inclined plane



43. A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F = \sqrt{x}$ as shown in figure. The value of x =___



44. If a body is placed on a rough inclined plane, the nature of forces acting on the body is (are)

(*i*) Gravitational force (*ii*) Electromagnetic force

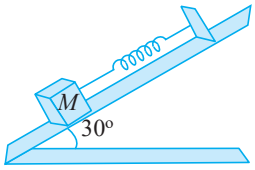
(*iii*) Nuclear force (*iv*) Weak nuclear force

(*a*) Only *i* (*b*) *i* and *iv*

(*c*) All (*d*) *i* and *ii*

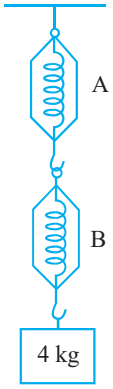
Spring Mass System

45. A body of mass 5 kg is suspended by a spring balance on an inclined plane as shown in figure. The spring balance measure:



(*a*) 50 N (*b*) 25 N (*c*) 500 N (*d*) 10 N

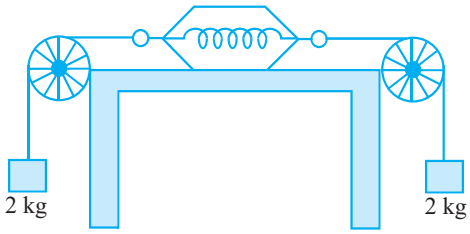
46. A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will read respectively:



(*a*) 4 kg and zero kg (*b*) zero kg and 4 kg

(*c*) 4 kg and 4 kg (*d*) 2 kg and 2 kg

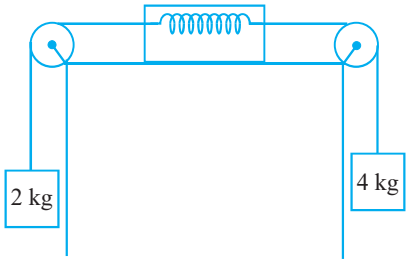
47. As shown in figure, two equal masses each of 2 kg are suspended from a spring balance. The reading of the spring balance will be



(*a*) Zero (*b*) 2 kg

(*c*) 4 kg (*d*) Between zero and 2 kg

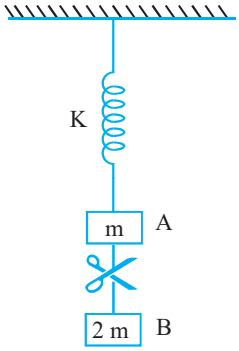
48. Reading of spring.



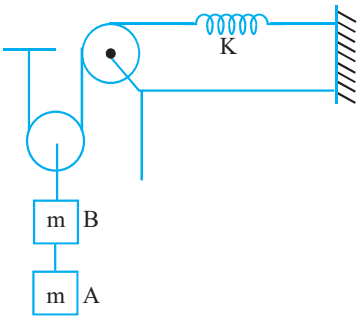
49. A spring whose unstretched length is l has a force constant k. The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be:

(*a*) $1/n^2$ (*b*) n^2 (*c*) $1/n$ (*d*) n

50. System as shown in figure is in equilibrium. If string is cut then find acceleration of A and B.



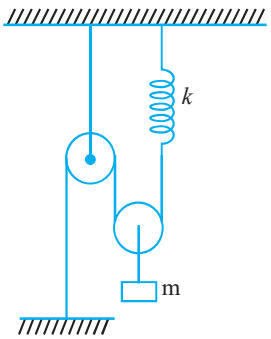
51. Initially both blocks A and B stay at equilibrium If string between A and B is cut, then acceleration of block A and B just after cut will be:



(*a*) g downward, g upward (*b*) g/2 downward, zero

(*c*) g downward, g downward (*d*) g/2 downward, g upward

52. A block of mass m shown in figure is in equilibrium. If it is displaced further by x and released. Find it acceleration just after it is released. Take pulleys to be light and smooth and strings light.



(*a*) $\frac{4kx}{5m}$ (*b*) $\frac{2kx}{5m}$

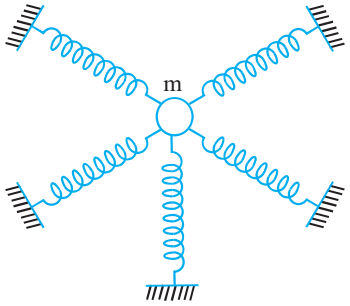
(*c*) $\frac{4kc}{m}$ (*d*) None of these

53. Three forces $F_1 = 10$ N, $F_2 = 8$ N, $F_3 = 6$N are acting on a particle of mass 5 kg. The forces F_2 and F_3 are applied perpendicularly, so that particle remains at rest. If the force F_1 is removed, then acceleration of particle is

(*a*) 4.8 m/s^2 (*b*) 7 m/s^2

(*c*) 2 m/s^2 (*d*) 0.5 m/s^2

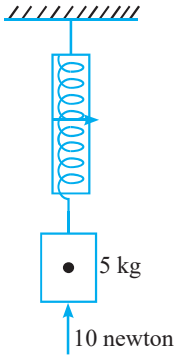
54. A sphere of mass m is kept in equilibrium with the help of several springs as shown in the figure. Measurement shows that one of the springs applies a Force $\vec{F}$ on the sphere. With what acceleration the sphere will move immediately after this particular spring is cut?



(*a*) Zero (*b*) $\vec{F}/m$

(*c*) $-\vec{F}/m$ (*d*) Insufficient information

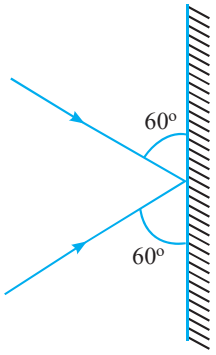
55. Reading of spring balance is ($g = 10\ m/s^2$) if object of mass 5 kg is in equilibrium,



(*a*) 10 N (*b*) 20 N (*c*) 30 N (*d*) 40 N

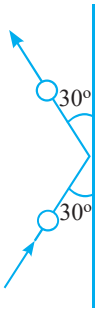
Newton's Second Law of Motion

56. A body of mass 3 kg moving with velocity 10 m/s hits a wall at an angle of 60° and returns at the same angle. The impact time was 0.2 s. Calculate the force exerted on the wall



(*a*) $150\sqrt{3}$N (*b*) $50\sqrt{3}$N (*c*) 100 N (*d*) 75 N

57. A 0.5 kg ball moving with a speed of 12 m/s strikes a hard wall at an angle of 30° with the wall. It is reflected with the same speed and at the same angle. If the ball is in contact with the wall for 0.25 s, the average force acting on the wall is:



(*a*) 48 N (*b*) 24 N

(*c*) 12 N (*d*) 96 N

58. A cricketer catches a ball of mass 150 g in 0.1 s moving with speed 20 m/s, then the experiences force of

(*a*) 300 N (*b*) 30 N

(*c*) 3 N (*d*) 0.3 N

59. A ball of mass 50 g is dropped from a height of 20 m. A boy on the ground hits the ball vertically upwards with a bat with an average force of 200 N, so that it attains a vertical height of 45 m. The time for which the ball remains in contact with the bat is [Take $g = 10\ m/s^2$]

(*a*) $1/20^{th}$ of a second (*b*) $1/40^{th}$ of a second

(*c*) $1/80^{th}$ of a second (*d*) $1/20^{th}$ of a second

60. A force of 6 N acts on a body at rest and of mass 1 kg. During this time, the body attains a velocity of 30 m/s. The time for which the force acts on the body is

(*a*) 7 second (*b*) 5 second

(*c*) 10 second (*d*) 8 second

61. In two different experiments, an object of mass 5 kg moving with a speed of 25 ms^{-1} hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively. Choose the correct option out of the following:

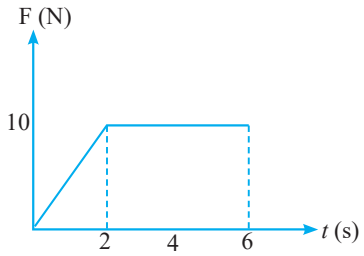
(*a*) Impulse and average force acting on the object will be same for both the cases.

(*b*) Impulse will be same for both the cases but the average force will be different.

(*c*) Average force will be same both the cases but the impulse will different.

(*d*) Average force and impulse will be different for both the cases.

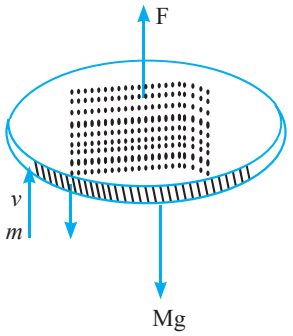
62. A body of mass 3 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is given by:



(*a*) Zero (*b*) 5 N-S

(*c*) 30 N-S (*d*) 50 N-S

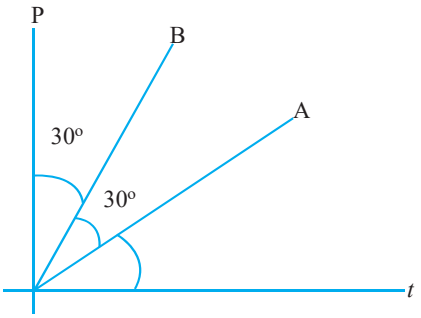
63. A disc of mass 1.0 kg kept floating horizontally in air by firing bullets of mass 0.05 kg each vertically at it, at the rate of 10 per second. If the bullets rebound with the same speed, the speed with which these are fired will be-



(*a*) 0.098 m/s (*b*) 0.98 m/s
(*c*) 9.8 m/s (*d*) 98.0 m/s

64. A stone of mass 1 kg is thrown with a velocity of 20 m/s across the frozen surface of a lake and it comes to rest after travelling a distance of 50 m. What is the magnitude of the force opposing the motion of the stone?

65. For two object P-t graph is given then find ratio of force acting on them.



66. Figures (*a*), (*b*), (*c*) and (*d*) show variation of force with time.

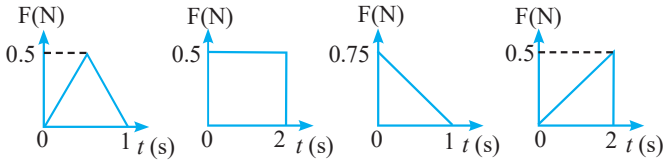


Fig. (*a*) Fig. (*b*) Fig. (*c*) Fig. (*d*)

The impulse is highest in figure

(*a*) Figure (*c*) (*b*) Figure (*d*)
(*c*) Figure (*a*) (*d*) Figure (*b*)

67. The momentum p (in kg/m) of a particle is varying with time t (in s) as $p = 2 + 3t^2$ The force acting on the particle at $t = 3$ s will be

(*a*) 18 N (*b*) 54 N (*c*) 9 N (*d*) 15 N

68. A force $\vec{F} = (2t\hat{i} + 3t^2\hat{j})$ N acts on an object moving in D plane. Find magnitude of change in momentum of the object in time interval t = 0 to t = 2s

69. At any instant the velocity of particle of mass 500 g is $(2t\hat{i} + 3t^2\hat{j})$ m/s. If the force acting on the particle at t = 1s is $(\hat{i} + x\hat{j})$ N. Then the value of x will be

(*a*) 2 (*b*) 4 (*c*) 6 (*d*) 3

70. If impulse varies with time t as (kg ms^{-1}) = $20t^2$ - 40t The change in momentum is minimum at

(*a*) t = 2s (*b*) t = 1s
(*c*) t = 1/2 s (*d*) t = 3 / 2s

Variable Mass System

71. Gravel is dropped onto a conveyer belt at a rate of 0.5 kg/s. The extra force required in newton to keep the belt moving at 2 m/s is:

(*a*) 1 N (*b*) 2 N
(*c*) 4 N (*d*) 0.5 N

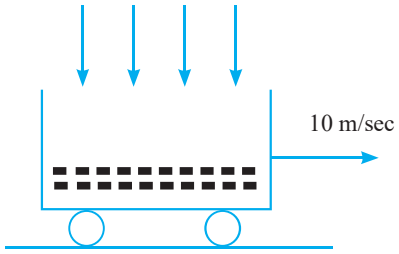
72. A balloon of mass 2 gram has a small hole pierced into it. The air comes out with a velocity of 4 m/s. If the balloon shrinks completely in 2.5 s. The average force acting on the balloon is

(*a*) 0.008 N (*b*) 0.0032 N
(*c*) 8 N (*d*) 3.2 N

73. A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on the balloon will be (in dyne).

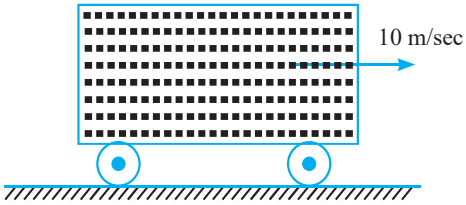
(*a*) 3 (*b*) 9
(*c*) 12 (*d*) 18

74. A cart of mass 1200 kg is moving with constant speed 10 m/sec. Due to heavy rain, water is entering in cart with rate 10 kg/sec, as shown in diagram then velocity of cart after 2 minute will be:



(*a*) 5 m/sec (*b*) 10 m/sec
(*c*) 7.5 m/sec (*d*) 2.5 m/sec

75. A cart having total mass 3000 kg is moving with constant speed 10 m/sec now sand is leaking from hole with rate 10 kg/sec, then speed of cart after 2 minute will be :



(*a*) 10 m/sec (*b*) 16.67 m/sec
(*c*) 15 m/sec (*d*) None

76. Water jet is coming out of a hose pipe of diameter 20 cm with speed 20 cm/sec. It strikes a man inelastically. Find the force exerted by water jet on the person.

(*a*) 5.04 N (*b*) 1.26 N
(*c*) 2.52 N (*d*) 9.4 N

77. A satellite in force free space sweeps stationary interplanetary dust at a rate $(dM/dt) = \alpha v$ Here v is the velocity. The acceleration of satellite of mass Mis M is

(*a*) $-2\alpha v^2/M$ (*b*) $-3\alpha v^2/M$

(*c*) $-\alpha v^2/M$ (*d*) $-\alpha v^2$

78. A cart is moving with a velocity 20 m/s. Sand is being dropped into the cart at the rate of 50 kg/min. The force required to move the card with constant velocity will be

(*a*) 50 N (*b*) 30.33 N

(*c*) 26.45 N (*d*) 16.66 N

79. A stream of water flowing horizontally with a speed of 15 ms^{-1} gushes out of a tube of cross-sectional area 10^{-2} m^2 and hits a vertical wall nearby. What is the force exerted on the wall by the impact of water, assuming it does not rebound?

(*a*) 2250 N (*b*) 2408 N

(*c*) 2048 N (*d*) None of these

Rocket Propulsion

80. A 800 kg rocket is fired from earth so that exhaust speed is 1200 m/s. Then calculate mass of fuel burning per second, to provide initial thrust to overcome its weight. ($g = 10m/(s^2)$)

81. A cracker rocket is ejecting gases at a rate of 0.05 kg/s with a velocity 400 m/s. The accelerating force on the rocket is:

(*a*) 20 dyns (*b*) 20 N

(*c*) 200 N (*d*) zero

82. If the force on a rocket, that releases the exhaust gases with a velocity of 300 m/s is 210 N, then the rate of combustion of the fuel is

(*a*) 0.07 kg/s (*b*) 1.4 kg/s

(*c*) 0.7 kg/s (*d*) 10.7 kg/s

83. A rocket of mass 5700 kg ejected mass at a constant rate of 15 kg/s with constant speed of 12 km/s. The acceleration of the rocket 1 minute after the blast is ($g = 10m/(s^2)$)

(*a*) 34.9 m/s^2 (*b*) 27.5 m/s^2

(*c*) 3.50 m/s^2 (*d*) 13.5 m/s^2

84. Suppose a rocket with an initial mass M_0, eject a mass Δm in the form of gases in time Δt, then the mass of the rocket after time t is:-

(*a*) $M_0 - \frac{\Delta m}{\Delta t}.t$ (*b*) $M_0 - \frac{\Delta m}{\Delta t}$ (*c*) $M_0 + \frac{\Delta m}{\Delta t}$ (*d*) M_0

Momentum & Conservation of Momentum

85. **Assertion (A):** Some force applied for some time causes the same change in momentum for different bodies

Reason (R): The total momentum of an isolated system of interacting bodies remains conserved

(*a*) If both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of the Assertion (A)

(*b*) If both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A)

(*c*) If Assertion (A) is correct but Reason (R) is incorrect

(*d*) If both Assertion (A) and Reason (R) are incorrect

86. A body of mass m is projected with initial speed u at an angle θ with the horizontal. The change in momentum of body after time t is:

(*a*) m u sin θ (*b*) 2 m u sin θ

(*c*) $m\,g\,t$ (*d*) Zero

87. A particle is acted upon by a force given by $F = (12t + 3t^2)N$ where t is in seconds. Find the change in momentum of that particle from $t = 1$ to $t = 3$ sec.

Gun-Bullet System

88. A bullet of mass 50 g is fired from a gun with initial velocity of 32 m/s. If mass of the gun is 4 kg, then calculate the recoil velocity of the gun.

89. A machine gun fires a bullet of mass 65 g with a velocity of 1300 m/s. The man holding it can exert a maximum force of 169 N on the gun. The number of bullets he can fire per second will be:

90. A bullet of mass 40 g is fired from a gun of mass 10 kg. If velocity of bullet is 400 m/s, the recoil velocity of the gun will be:

(*a*) 1.6 m/s in the direction of bullet

(*b*) 1.6 m/s opposite to the direction of bullet

(*c*) 1.8 m/s in the direction of bullet

(*d*) 1.8 m/s opposite to the direction of bullet

91. **Assertion (A):** A bullet is fired from a rifle. If the rifle recoils freely, the kinetic energy of rifle is more than that of bullet

Reason (R): Kinetic energy depends on mass directly by formula $k.e = \frac{1}{2}mv^2$

(*a*) If both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of the Assertion (A)

(*b*) If both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A)

(*c*) If Assertion (A) is incorrect but Reason (R) is correct

(*d*) If both Assertion (A) and Reason (R) are incorrect

92. A machine gun fire bullets of mass 40 gm each with the velocity of 100 m/sec. If average force is exerted on the gun F is 400 N, then find the no. of bullets fired per minutes.

Newton's third Law of Motion

93. **Assertion (A):** A block of Table is at rest because gravitational force and normal Rx are action Rx^n pair and cancel out each other

Reason (R): Newton 3rd law states that every action has equal and opposite reaction in opposite direction

(*a*) If both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of the Assertion (A)

(*b*) If both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A)

(*c*) If Assertion (A) is incorrect but Reason (R) is correct

(*d*) If both Assertion (A) and Reason (R) are incorrect

94. Swimming is possible on account of

(*a*) First law of motion

(*b*) Second law of motion

(*c*) Third law of motion

(*d*) Newton's law of gravitation

95. You are on a frictionless horizontal plane. How can you get off if no horizontal force is exerted by pushing against the surface

(*a*) By jumping

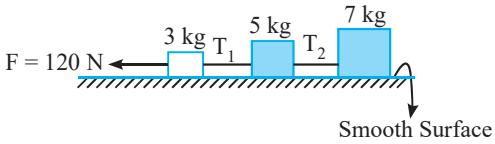
(*b*) By spitting or sneezing

(*c*) By rolling your body on the surface

(*d*) By running on the plane

Connected Body Motion

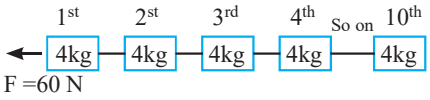
96. Find T_1 and T_2 in the given figure are



(*a*) 28 N, 48 N (*b*) 48 N, 28 N

(*c*) 96 N, 56 N (*d*) 56 N, 96 N

97. Find tension b/w 7th & 8th block

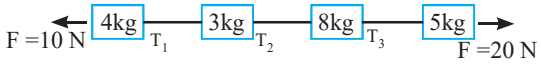


98. A block of mass *M* is pulled along a horizontal frictionless surface by rope of mass *m*. If a force *P* is applied at the free end of the rope, the force exerted by the rope on block is

(*a*) $\frac{Pm}{M+m}$ (*b*) $\frac{Pm}{M-m}$

(*c*) P (*d*) $\frac{PM}{M+m}$

99. Find tension T_1, T_2 & T_3

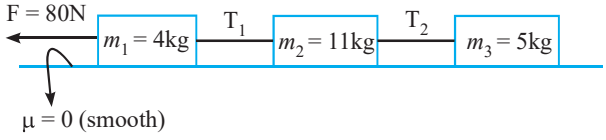


100. A train has 10 wagons each of mass 1000 kg attached to it. They are being pulled by force 10^4 N. Find out force exerted on last four wagons.

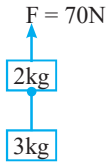
(*a*) 4×10^3 N (*b*) 4×10^4 N

(*c*) 5×10^4 N (*d*) 5×10^3 N

101. Find tension (T_1 & T_2) and acceleration of the system



102. Find tension in string connected between 2 kg and 3 kg.



103. A monkey of mass 50 kg climbs on a rope which can withstand the tension of 350 N. If monkey initially climbs down with acceleration of 4m / (s^2) and then climbs up with acceleration of 5 m/(s^2) . Choose correct option

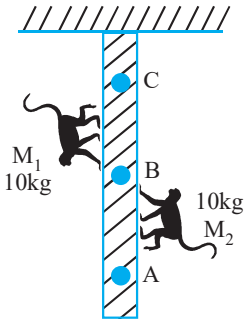
(*a*) T = 700 N while climbing upward

(*b*) T = 350 while going downward

(*c*) Rope will break while climbing upward

(*d*) Rope will break while going downward

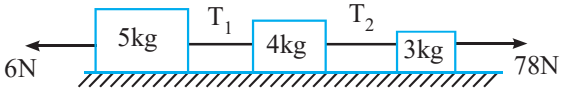
104. If monkey M_1 is at rest and breaking strength of string is 260 N, then a max of M_2 with which it can accelerate, so that string will not break is:



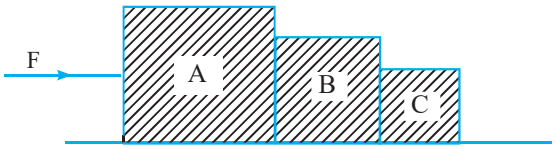
(*a*) 8 m/sec^2↑ (*b*) 6 m/sec^2↑

(*c*) 4 m/sec^2↑ (*d*) None

105. What will be the tension T_1 and T_2 in the given figure?



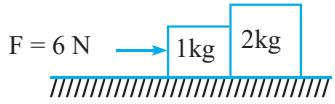
106. Three blocks A, B and C of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a frictionless surface, as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between A and B is



(*a*) 18 N (*b*) 2 N

(*c*) 6 N (*d*) 8 N

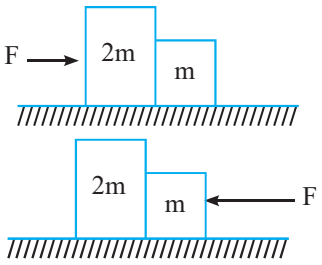
107. Arrangement of two block system is as shown. The net force acting on1 kg and 2 kg blocks are (assuming the surface to the frictionless) respectively



(*a*) 4 N, 8N (*b*) 1 N, 2N

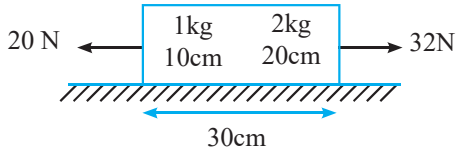
(*c*) 2 N, 4 n (*d*) 3 N, 6 N

108. Two blocks are in contact on a frictionless table. One has mass m and the other 2m. A force F is applied on 2m as shown in the figure. Now the same force F is applied from the right on m. In the two cases respectively, the ratio of force of contact between the two blocks will be



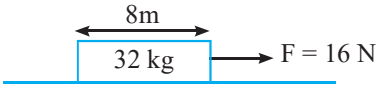
(*a*) Same (*b*) 1 : 2
(*c*) 2 : 1 (*d*) 1 : 3

109. Figure shows a uniform rod of length 30 cm having a mass 3.0 kg. The rod is pulled by constant forces of 20 N and 32 N as shown. Find the force exerted by 20 cm part of the rod on the 10 cm part (all surfaces are smooth) is:

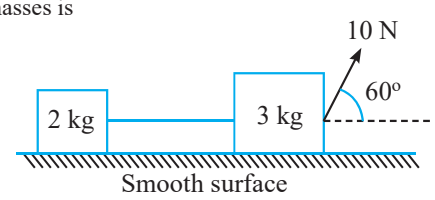


(*a*) 36 N (*b*) 12N
(*c*) 64 N (*d*) 24 N

110. A massive string of length 8 m and mass 32 kg, then find tension of a point 3 m away from on end where force is applied.

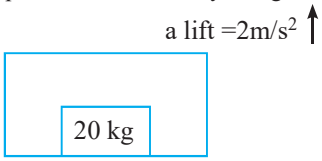


111. Figure shows two blocks connected by a light inextensible string as shown in figure. A force of 10 N is applied on the bigger block at 60° with horizontal, then the tension in the string connecting the two masses is



(*a*) 5 N (*b*) 2 N
(*c*) 1 N (*d*) 3 N

112. Find force applied on base of lift by 20 kg block.



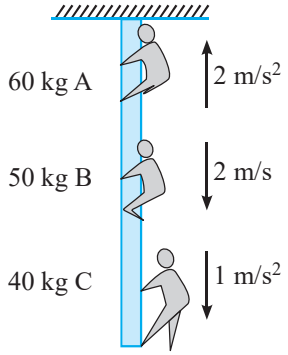
113. A lift of mass 1000 kg is moving with acceleration of 1 m/s² in upward direction, then the tension developed in string which is connected to lift is

(*a*) 9800 N (*b*) 10,800 N
(*c*) 11,000 N (*d*) 10, 000 N

114. A mass of 1 kg is suspended by a thread. It is (i) lifted up with an acceleration 4.9 m/s², (ii) lowered with an acceleration 4.9 m/s². The ratio of the tensions is

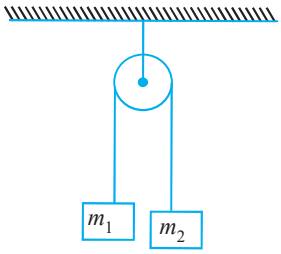
(*a*) 1 : 3 (*b*) 1 : 2 (*c*) 3 : 1 (*d*) 2 : 1

115. Tension in the rope at the rigid support is $g = 10 m / s^2$

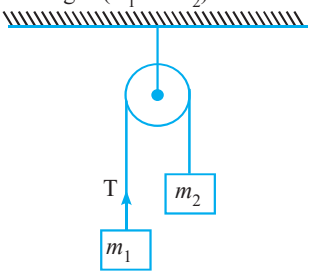


(*a*) 760 N (*b*) 1360 N (*c*) 1580 N (*d*) 1620 N

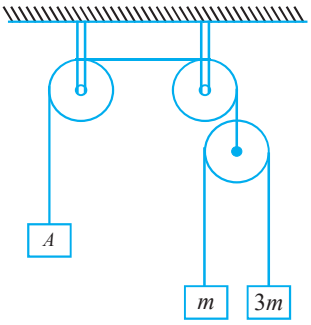
116. If $(m_2 > m_1)$ then find acceleration of m_1 and m_2 and Tension in string.



117. Find tension in string if $(m_1 >> m_2)$

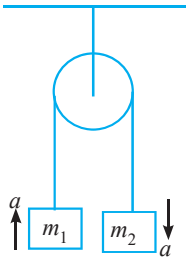


118. In the given figure, find mass of the block A if it remains at rest, when the system is released form rest. Pulleys and strings are massless. [g = 10 m/s²]

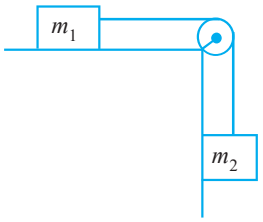


(*a*) m (*b*) 2 m (*c*) 2.5 m (*d*) 3 m

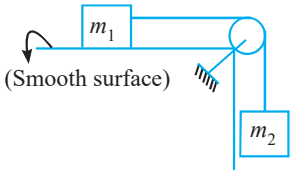
119. Find acceleration of m_1 and m_2 in given pulley block syst $(m_2 > m_1)$



120. Find acceleration of Block m_1 and m_2 and tension in string where horizontal surface is smooth.

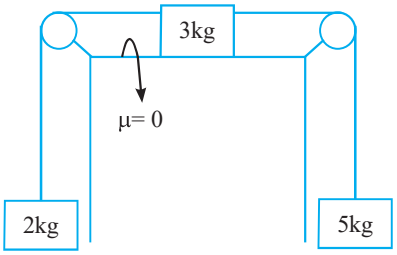


121. If $m_1 = m_2 = 1$ kg then find distance travelled by m_1 in first $\frac{1}{2}$ sec.

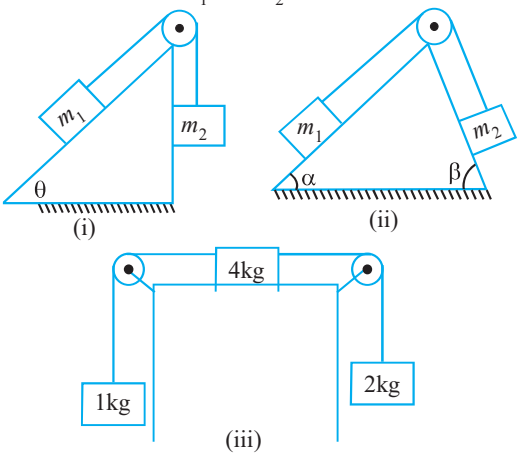


(*a*) 0.625 m (*b*) 0.425 m (*c*) 0.725 m (*d*) 0.525 m

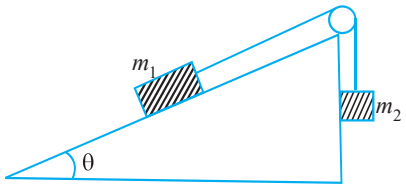
122. Find acceleration of system.



123. Find acceleration on M_1 and M_2

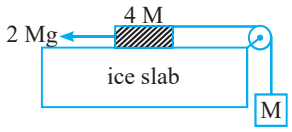


124. Two bodies of masses m_1 = 5 kg and m_2 = 3 kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be: [Take g = 10 ms^{-2}]

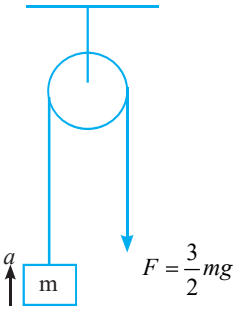


(*a*) 30 N (*b*) 40 N (*c*) 50 N (*d*) 60 N

125. A hanging mass M is connected to a four times bigger mass by using a string pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is x/5 Mg for x =_____Neglect mass of the string and friction of the block (bigger mass) with ice slab

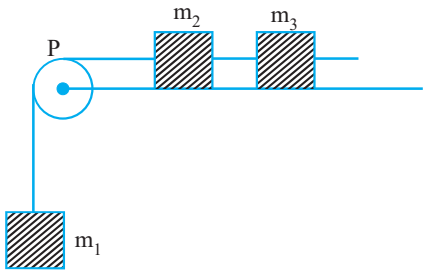


126. In the arrangement shown, the mass m will ascend with an acceleration (Pulley and rope are massless)



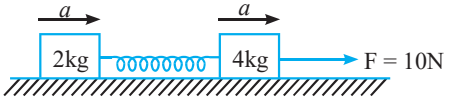
(*a*) Zero (*b*) g/2 (*c*) g (*d*) 2g

127. A system consists of three masses m_1, m_2 and m_3 connected by a string passing over a pulley P. The mass m_1 hangs freely and m_2 and m_3 are on a rough horizontal table (The coefficient of friction = μ). The pulley is frictionless and of negligible mass. The downward acceleration of mass m_1 is: (Assume $m_1 = m_2 = m_3 = m$)



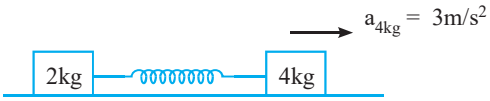
(*a*) $\frac{g(1-g\mu)}{9}$ (*b*) $\frac{2g\mu}{3}$ (*c*) $\frac{g(1-2\mu)}{3}$ (*d*) $\frac{g(1-2\mu)}{2}$

128. Two blocks of mass 2 kg and 4 kg are accelerated with same acceleration by a force 10 N as shown in figure on a smooth horizontal surface. Then the spring force between the two blocks will be (spring is massless)

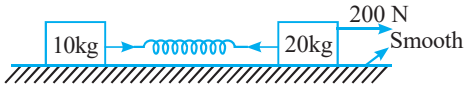


(*a*) 5 N (*b*) 10 N (*c*) $\frac{10}{3}N$ (*d*) $\frac{5}{3}N$

129. Find acceleration of 2 kg. where acceleration of 4kg is 3m/s^2 in forward direction.

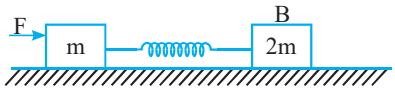


130. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass. At the instant shown the 10 kg mass has acceleration 12 m/s^2 towards right. The acceleration of 20 kg mass at this instant is:

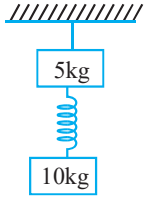


(*a*) 12 m/s^2 (*b*) 4 m/s^2
(*c*) 10 m/s^2 (*d*) zero

131. Find acceleration of B block if acceleration of a block is a_0 towards left.

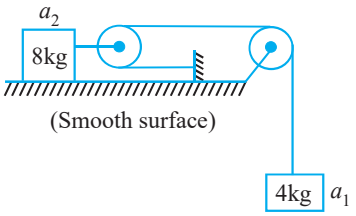


132. System in figure is in equilibrium. Now string is cut. Acceleration of both block will be:



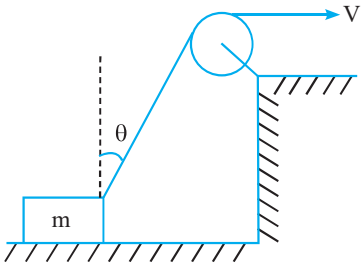
(*a*) g↓,0 (*b*) 3g↓,0 (*c*) 2g↓, g↓ (*d*) 3g↓, g↓

133. If pulleys shown in the diagram are smooth and massless and a_1 and a_2 are acceleration of blocks of mass 4 kg and 8 kg respectively, then



(*a*) $a_1 = a_2$ (*b*) $a_1 = 2a_2$ (*c*) $2a_1 = a_2$ (*d*) $a_1 = 4a_2$

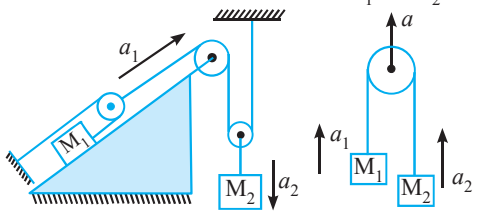
134. A block is dragged on smooth plane with the help of a rope which moves with velocity *v*. The horizontal velocity of the block is:



(*a*) v (*b*) $\frac{v}{\sin\theta}$

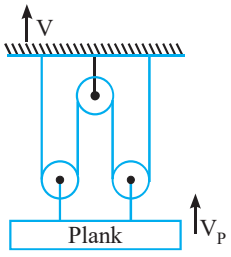
(*c*) $v \sin\theta$ (*d*) $\frac{v}{\cos\theta}$

135. Constrain Motion: Find relation between a_1 and a_2

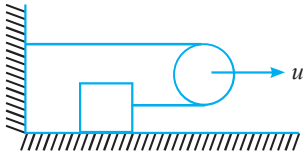


(i) (ii)

136. Constrain Motion: Find relation between V and V_p



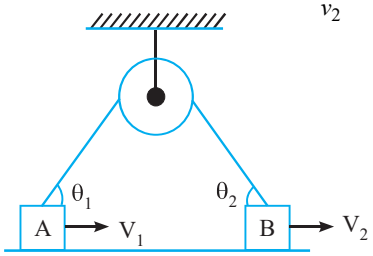
137. In the figure shown, the pulley is moving with velocity *u*. The velocity of the block attached with string :



(*a*) 4 *u* (*b*) 3 *u*

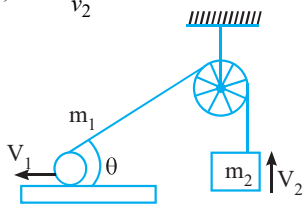
(*c*) 2 *u* (*d*) *u*

138. In the figure shown, blocks A and B move with velocities V_1 and V_2 along horizontal direction. The ratio of $\frac{v_1}{v_2}$:



(*a*) $\frac{\sin\theta_2}{\sin\theta_1}$ (*b*) $\frac{\sin\theta_1}{\sin\theta_2}$ (*c*) $\frac{\cos\theta_2}{\cos\theta_1}$ (*d*) $\frac{\cos\theta_1}{\cos\theta_2}$

139. In figure, a ball of mass m_1 and a block of mass m_2 are joined together with an inextensible string. The ball can slide on a smooth horizontal surface. If v_1 and v_2 are the respective speeds of the ball and the block, Find $\frac{v_1}{v_2}$



(*a*) cos θ (*b*) sec θ

(*c*) tan θ (*d*) sin θ

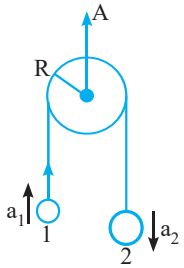
140. Constrain Motion : Find relation between $a_1$and a_2



(*a*) $a_1 = a_2$ (*b*) $a_1 = 3a_2$

(*c*) $a_1 = 2a_2$ (*d*) $2a_1 = 3a_2$

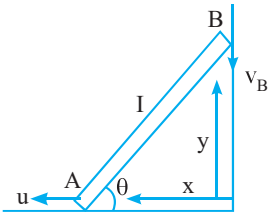
141. Two masses are connected by a string which passes over a pulley accelerating upward at a rate A as shown. If a_1 and a_2 be the acceleration of bodies 1 and 2 respectively then :



(*a*) $A = a_1 - a_2$ (*b*) $A = a_1 + a_2$

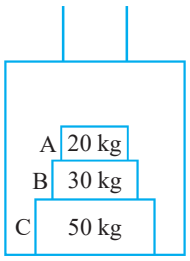
(*c*) $A = \frac{a_1 - a_2}{2}$ (*d*) $A = \frac{\alpha_1 + a_2}{2}$

142. Figure shows a rod of length l resting on a wall and the floor. Its lower end A is pulled towards left with a constant velocity u. As a result of this, end B starts moving down along the wall. Find the velocity of the other end B downward when rod makes an angle θ with the horizontal.



(*a*) $2u \sin\theta$ (*b*) $u \sin\theta$ (*c*) $u \cot\theta$ (*d*) $2u \sin\theta$

143. In shown situation elevator is moving upward with acceleration of 5 m/s^2



Column-I		Column-II	
1	Net force acting on B	P	150 N
2	Normal reaction between A and B	Q	300 N
3	Normal reaction between B and C	R	450 N
4	Normal reaction between C and elevator	S	750 N
5		T	1500 N

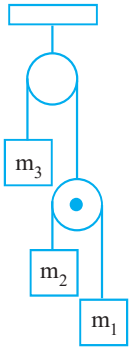
(*a*) (1) – P, (2) – Q, (3) – S, (4) – T

(*b*) (1) – Q, (2) – Q, (3) – R, (4) – T

(*c*) (1) – P, (2) – R, (3) – S, (4) – S

(*d*) (1) – Q, (2) – R, (3) – R, (4) – T

144. Three masses m_1, m_2 and m_3 are attached to a string-pulley system as shown. All the three masses are held at rest and then released. To keep m_3 at rest, the condition is



(*a*) $m_3 = m_1 + m_2$ (*b*) $m_3 = \dfrac{2m_1m_2}{m_1+m_2}$

(*c*) $m_3 = \sqrt{m_1m_2}$ (*d*) $m_3 = \dfrac{4m_1m_2}{m_1+m_2}$

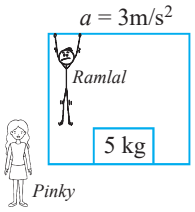
145. With what minimum acceleration can a fire man slide down a rope whose breaking strength is 3/4th of his weight.

(*a*) g/4 m/s^2 (*b*) g m/s

(*c*) 3/4m/s^2 (*d*) Zero

Pseudo Force

146. Find contact force between block and surface of lift. w.r.t Ramal & Pinky



147. Normal on block with respect to pinky & Ramlal will be

(*a*) Same (*b*) Different

(*c*) Don't know

148. **Statement I:** An elevator can go up on down with uniform speed when its weight is balanced with the tension of its cable.

Statement II: Force exerted by floor of an elevator on foot of person standing on it is more than his/her weight when elevator goes down with increasing speed

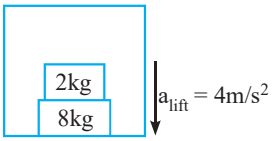
(*a*) Statement I and Statement II is false

(*b*) Statement I and Statement II is true

(*c*) Statement I is false but statement II is true

(*d*) Statement I is true but statement II is false

149. Find contact force between 2 kg and 8 kg

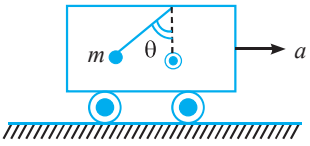


150. A small metallic sphere of mass m is suspended from the ceiling of a car accelerating on a horizontal road with constant acceleration a. The tension in the string attached with metallic sphere is

(*a*) Mg (*b*) $m(g + a)$

(*c*) $m(g - a)$ (*d*) $m\sqrt{g^2 + a^2}$

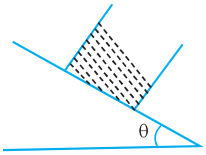
151. If trolley accelerates horizontally with acceleration a then bob is displaced backward from its initial vertical position. The angular deflection of the bob in equilibrium is

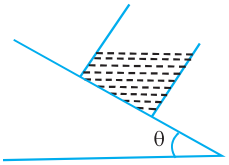


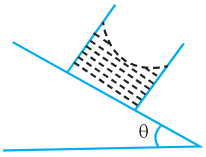
(*a*) $\theta = \cos^{-1}\left(\dfrac{a}{g}\right)$ (*b*) $\theta = \sin^{-1}\left(\dfrac{a}{g}\right)$

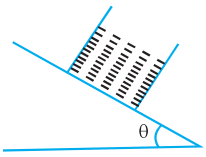
(*c*) $\theta = \cot^{-1}\left(\dfrac{a}{g}\right)$ (*d*) $\theta = \tan^{-1}\left(\dfrac{a}{g}\right)$

152. A beaker is half filled with water. It is allowed to slip down on smooth inclined plane with angle of inclination θ to the horizontal. The level of water in the beaker will be :-

(a) 

(b) 

(c) 

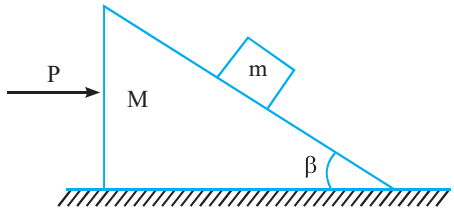
(d) 

153. A block of mass m kg is kept on a weighing machine in an elevator. If the elevator is retarding upward by a ms^{-2}, the reading of weighing machine is (in kg)

(a) mg (b) $m(g-a)$

(c) $m\left(1-\frac{a}{g}\right)$ (d) $m(g+a)$

154. A block of mass m, is kept on a wedge of mass M, as shown in figure such that mass m remains stationary w.r.t. wedge. The magnitude of force P is



(a) g tan β (b) mg tan β

(c) (m + M)g tan β (d) mg cot β

155. A block of mass m is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (g is acceleration due to gravity) will be

(a) mg cos θ (b) mg sin θ (c) mg (d) mg/cos θ

156. Select the correct statement regarding pseudo force

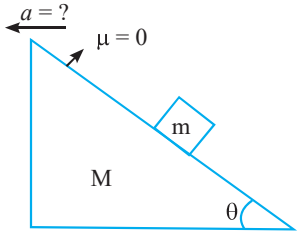
(a) It is electromagnetic in origin

(b) Newton's 3rd law is applicable for it

(c) It is a fundamental force

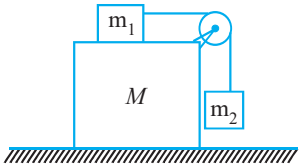
(d) It is used to make Newton's law applicable in non-inertial frame

157. Find acceleration of inclined plane so that block will fall freely.



(a) g tan θ (b) g cot θ (c) g (d) g sin θ

158. In the given arrangement all surfaces are smooth. What acceleration should be given to the system, for which the block m_2 doesn't slide down?



(a) $\frac{m_2 g}{m_1}$ (b) $\frac{m_1 g}{m_2}$ (c) g (d) $\frac{m_2 g}{m_1+m_2}$

159. A monkey of mass 20 kg is holding a vertical rope. The rope will not break when a mass of 25 kg is suspended from it but will break if the mass exceeds 25 kg. What is the maximum acceleration with which the monkey can climb up along the rope ? (g = 10m / (s^2)

(a) 5 m/s^2 (b) 10 m/s^2 (c) 25 m/s^2 (d) 2.5 m/s^2

160. A man of mass m is standing in an elevator moving downward with an acceleration. $\frac{g}{4}$ The force exerted by the bottom surface of the elevator on the man will be

(a) $\frac{3mg}{4}$ (b) $\frac{mg}{4}$ (c) $\frac{5mg}{4}$ (d) $\frac{7mg}{4}$

161. Two blocks of masses 2 kg and 4 kg are hanging with the help of massless string passing over an ideal pulley inside an elevator. The elevator is moving upward with an acceleration. $\frac{g}{2}$ The tension in the string connected between the blocks 2 will be (Take g = 10m / (s^2)

(a) 40 N (b) 60 N (c) 80 N (d) 20 N

162. A reference frame attached to the earth-

(A) Is an inertial frame by definition

(B) Cannot be an inertial frame because the earth is revolving round the sun

(C) Is an inertial frame because Newton's laws are applicable in this frame

(D) Cannot be an inertial frame because the earth is rotating about its axis

(a) (A, B) (b) (B, C) (c) (A, C) (d) (B, D)

Apparent Weight

163. A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 5 m/s^2. What would be the reading on the scale? (g = 10m /s^2)

(a) Zero (b) 400 N (c) 800 N (d) 1200 N

164. A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be:

(a) g/4 (b) g/2 (c) 3g/4 (d) g

165. A man of mass 90 kg is standing in an elevator whose cable broke suddenly. If the elevator falls, apparent weight of the man is:

(a) 90 N (b) 90 g N

(c) Zero (d) any negative value

166. A person is standing in an elevator. In which situation he finds his weight less

(a) When the elevator moves upward with constant acceleration

(b) When the elevator moves downward with constant acceleration

(c) When the elevator moves upward with uniform velocity

(d) When the elevator moves downward with uniform velocity

167. A bird is sitting in a large closed cage which is placed on a spring balance. It records a weight of 25 N. The bird (mass $m = 0.5$kg) flies upward in the cage with an acceleration of $2\text{m}/\text{s}^2$) The spring balance will now record a weight of

(*a*) 24 N (*b*) 25 N (*c*) 26 N (*d*) 27 N

Friction on Horizontal & Vertical Surface

168. The limiting friction between two bodies in contact is independent of

(*a*) Nature of the surface in contact

(*b*) The area of surfaces in contact

(*c*) Normal reaction between the surfaces

(*d*) The materials of the bodies

169. Which of the following is self-adjusting force?

(*a*) Static friction (*b*) Limiting friction

(*c*) Kinetic friction (*d*) Rolling friction

170. Maximum force of friction is called

(*a*) Limiting friction (*d*) Static friction

(*c*) Sliding friction (*d*) Rolling friction

171. Which is a suitable method to decrease friction?

(*a*) Polishing (*b*) Lubrication

(*c*) Ball bearing (*d*) All of these

172. Given below are two statements:

Statement I: The limiting force of static friction depends on area of contact and independent of materials independent

Statement II: The limiting force of kinetic friction is independent of area of contact and depends on materials

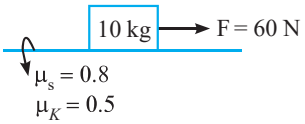
(*a*) Statement I is incorrect but statement II is correct

(*b*) Statement I and Statement II is correct

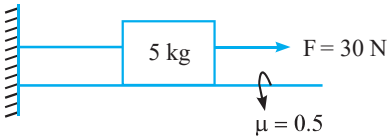
(*c*) Statement I and Statement II is incorrect

(*d*) Statement I is correct but statement II is incorrect

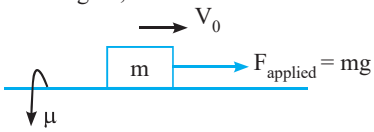
173. Find acceleration and friction between block and ground.



174. Find tension in string .....

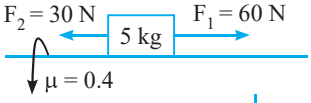


175. Object is moving with constant velocity under the applied force mg as shown in figure, then find contact force.

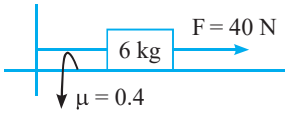


(*a*) mg (*b*) 2 mg (*c*) μmg (*d*) $\sqrt{2}mg$

176. Find acceleration and friction.



177. Find Tension in string.....



178. Object is thrown with velocity V_0 on rough surface of coefficient of friction μ, then stopping distance and time,

179. A body of mass 10 kg is moving with an initial speed of 20 m/s. The body stops after 5s due to friction between body and the floor. The value of the coefficient of friction is: (Take acceleration due to gravity $g = 10\text{ms}^{-2}$)

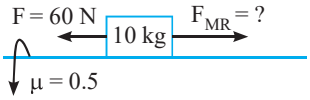
(*a*) 0.2 (*b*) 0.3 (*c*) 0.5 (*d*) 0.4

180. An object of mass 1 kg moving on a horizontal surface with initial velocity 8 m/s comes to rest after 10 s. If one wants to keep the object moving on the same surface with velocity 8 m/s the force required is

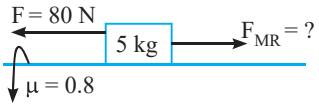
(*a*) 0.4 N (*b*) 0.8 N

(*c*) 1.2 N (*d*) Zero

181. Find the applied force F_{MR} to keep the object at rest.



182. Find value of F_{MR} to keep the object at rest.



183. A heavy box is slid across a rough floor with an initial speed of 4 m/s It stoops moving after 8 seconds. If the average resisting force of friction is 10 N, the mass of the box (in kg) is:

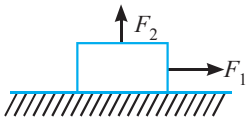
(*a*) 40 (*b*) 20 (*c*) 5 (*d*) 2.5

184. A child weighing 25 kg slides down a rope hanging from a branch of a tall tree. If the force of friction acting against him is 200 N, the acceleration of child is ($g = 10\text{m}/\text{s}^2$)

(*a*) 22.5 m/s^2 (*b*) 8 m/s^2

(*c*) 5 m/s^2 (*d*) 2 m/s^2

185. In the figure shown, horizontal force F_1 is applied on a block but the block does not slide. Then as the magnitude of vertical force F_2 is increased from zero the block begins to slide; the correct statement is



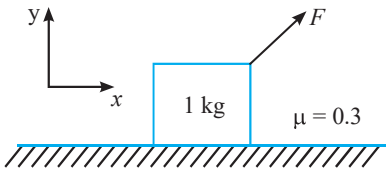
(*a*) The magnitude of normal reaction on block increases

(*b*) Static frictional force acting on the block increases

(*c*) Maximum value of static frictional force decreases

(*d*) All of these

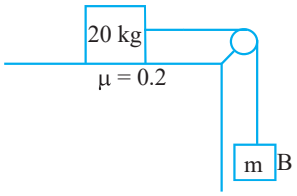
186. A force $\vec{F} = \hat{i} + 4\hat{j}$ acts on the block shown. The force of friction acting on the block is



(*a*) $-\hat{i}$ (*b*) $-18\hat{i}$

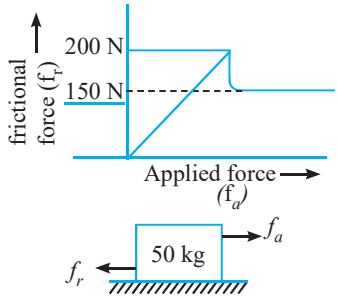
(*c*) $-2.4\hat{i}$ (*d*) $-3\hat{i}$

187. In the figure shown, the coefficient of static friction between the block A of mass 20 kg and horizontal table is 0.2. What should be the minimum mass of hanging block just beyond which blocks start moving?



(*a*) 2 kg (*b*) 3 kg (*c*) 4 kg (*d*) 5 kg

188. The graph between applied force and frictional force for a body of mass 50 kg placed in a horizontal surface is as shown in figure the value of coefficient of static and kinetic friction will be

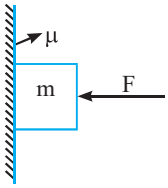


(*a*) 0.4, 0.3 (*b*) 0.3, 0.2 (*c*) 0.4, 0.2 (*d*) 0.2, 0.3

189. A metallic chain 1 long lies on a horizontal surface of a table. The chain starts sliding on the table if 25 cm (or more of it) hangs over the edge of a table. The correct value of the coefficient of friction between the table and the chain is

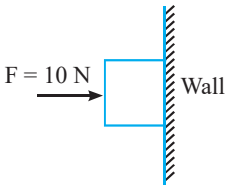
(*a*) $\frac{1}{3}$ (*b*) $\frac{2}{3}$ (*c*) $\frac{1}{4}$ (*d*) $\frac{1}{5}$

190. If object is at rest then find friction force acting on object.



(*a*) μmg (*b*) μF (*c*) mg (*d*) Can't say

191. A horizontal force 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2, the weight of the block is



(*a*) 20 N (*b*) 50 N (*c*) 100 N (*d*) 2 N

192. A block of mass 5 kg is placed at rest on a table of rough surface. Now, if a force of 30 N is applied in the direction parallel to surface of table, the block slides through a distance of 50*m* in an interval of time 10*s*. Coefficient of kinetic friction is

(*a*) 0.25 (*b*) 0.75 (*c*) 0.6 (*d*) 0.5

193. A Bag is gently dropped on a conveyer belt moving at speed of 2 m/s. The coefficient of friction between the conveyer belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by bag on belt during slipping motion, is

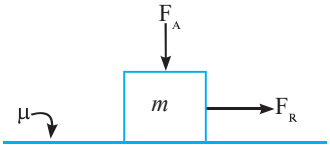
(*a*) 2 *m* (*b*) 0.5 *m* (*c*) 3.2 *m* (*d*) 0.8 *m*

194. A block of mass *m* is placed on a flat horizontal surface and the coefficient of friction between the block and surface is μ. A force F_1 is applied to block from above and a force F_R is applied to right. In all situations being considered below the block remains at rest. Let *f* be the force of friction on the block

(P) *f* increases if *m* is increased

(Q) *f* increases if F_A is increased

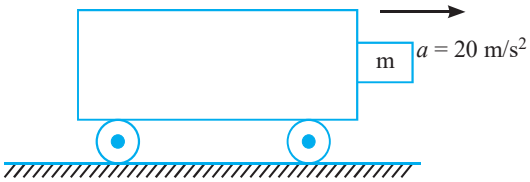
(R) *f* increases if F_R is increased



(*a*) Only P is true (*b*) Only Q is true

(*c*) P, Q is true (*d*) Only R is true

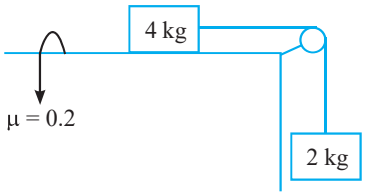
195. A vehicle is moving on a road with an acceleration $a = 20 m/s^2$) as shown in figure. The frictional coefficient between the block of mass (*m*) and the vehicle so that block is does not fall downwards is ($g = 10 m/s^2$)



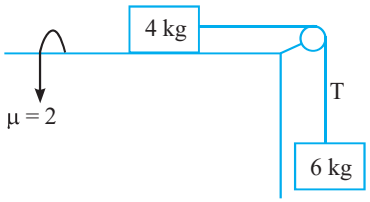
(*a*) 0.5 (*b*) 0.4

(*c*) 2 (*d*) 0.7

196. Two blocks of masses 4 kg and 2 kg are connected by a light inextensible string passing over a smooth pulley. The 4 kg block is placed on a horizontal surface with a coefficient of friction μ = 0.2, while the 2 kg block is hanging vertically. The system is released from rest. Find friction, tension in string and acceleration of system.



197. Find tension, friction and acceleration of system.

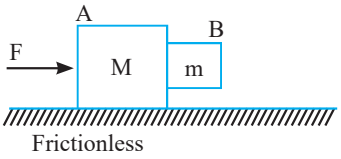


198. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μk. When the block A is sliding on the table, the tension in the string is

(*a*) $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$ (*b*) $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$

(*c*) $\frac{(m_1 m_2(1 + \mu_k)g}{(m_1 + m_2)}$ (*d*) $\frac{(m_1 m_2(1 - \mu_k)g}{(m_1 + m_2)}$

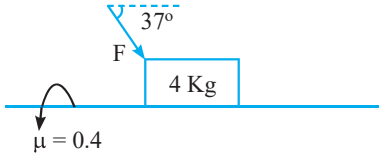
199. Coefficient of friction between A and B is μ. The minimum force F with which A will be pushed such that B will not slip down is



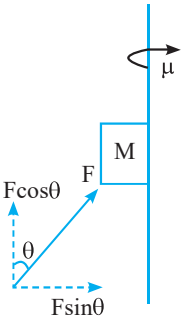
(a) $\frac{mg}{\mu}$ (b) $\frac{mg}{\mu}$

(c) $\frac{(M+m)g}{\mu}$ (d) $\frac{(M-m)g}{\mu}$

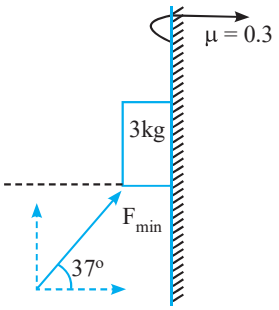
200. Find value of F so that object can move.



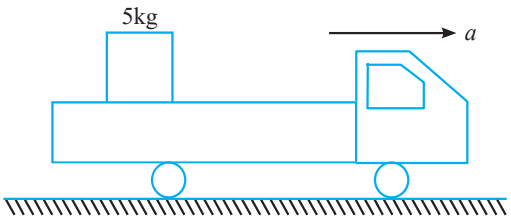
201. Find minimum value of *F* to keep the object at rest.



202. A block of mass 3 kg is held stationary against a vertical wall by applying a force F_{min} at an angle of 37° to the horizontal. The coefficient of friction between the block and the wall is μ = 0.3. Determine the minimum force required to keep the block at rest.



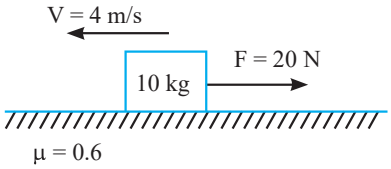
203. What will be the maximum acceleration of the truck so that the block of mass 5 kg placed on the truck does not slip? If the coefficient of friction mu =0.3:



(a) 3m/s² (b) 2m/s²

(c) 1m/s² (d) Zero

204. A block of mass 10 kg is moving on a rough surface as shown in figure. the frictional force acting on block is:- (g = 10m/s²)



(a) 20 N (b) 60 N

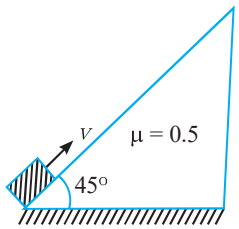
(c) 40 N (d) 80 N

Friction on inclined Plane

205. A cubical block rests on a plane of $\mu = \sqrt{3}$. The angle through which the plane be inclined to the horizontal so that the block just slides down will be

(a) 30° (b) 40° (c) 60° (d) 75°

206. A block of mass 1 kg is projected from the lowest point up along the inclined plane. If g = 10 ms⁻², the retardation experienced by the block is



(a) $\frac{15}{\sqrt{2}} ms^{-2}$ (b) $\frac{5}{\sqrt{2}}$ ms⁻² (c) $\frac{10}{\sqrt{2}}$ ms⁻² (d) Zero

207. if a block moving up an inclined plane at 30° with a velocity of 5m/s stops after 0.5 s, then coefficient of friction will be nearly

(a) 0.5 (b) 0.6 (c) 0.9 (d) 1.1

208. A block of mass m slides down an inclined plane inclined at angle 30° with an acceleration $\frac{g}{4}$. The value of coefficient of kinetic friction will be

(a) $\frac{2\sqrt{3}+1}{2}$ (b) $\frac{1}{2\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2\sqrt{3}-1}{2}$

209. A block of mass 10 kg is released on rough incline plane. Angle of inclination is 30°. Block start descending with acceleration 2 m/s². Kinetic friction force acting on block is (take g = 10m /s²)

(a) 10 N (b) 30 N

(c) 50 N (d) 50√3N

210. Block of mass 10 kg is moving on inclined plane at angle 37° with constant velocity 10 m/s. The coefficient of kinetic friction between incline plane and block is

(a) 0.57 (b) 0.75

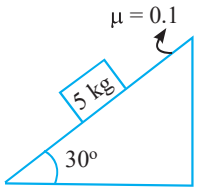
(c) 0.5 (d) None of these

211. A block (mass = M kg) is placed on a rough inclined plane. A force F is applied parallel to the inclined (as shown in figure) such that it just starts moving upward. The value of F is if angle of inclination is 37°

(a) Mg sinθ – μ Mg cosμ (b) Mg sinθ + μMg cosθ

(c) Mg sinθ (d) μ Mg cosθ

212. A block of mass 5 kg is placed on a rough inclined surface as shown in the figure. If $\vec{F}_1$ is the force required to just move the block up the inclined plane and $\vec{F}_2$ is the force required to just prevent the block from sliding down, then the value of $|\vec{F}_1| - |\vec{F}_2|$ is



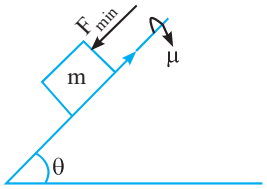
(a) $5\sqrt{3}$ N (b) 10 N (c) $\frac{5\sqrt{3}}{2}$ N (d) $25\sqrt{3}$ N

213. A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be:

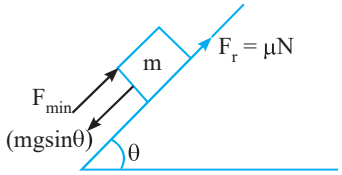
(a) Mg (b) Mg cos ω

(c) $\sqrt{Mg\sin\theta + Mg\cos\theta}$ (d) $Mg\sin\theta\sqrt{1+\mu}$

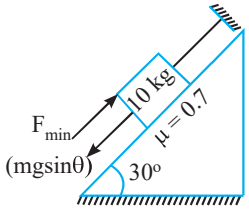
214. If μ > tan θ, then find minimum force required to move this object.



215. If μ < tanθ, then find force required to keep the object at rest.



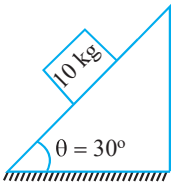
216. The tension T in the string shown in figure is



(a) Zero (b) 50 N

(c) 35 N (d) 3 N

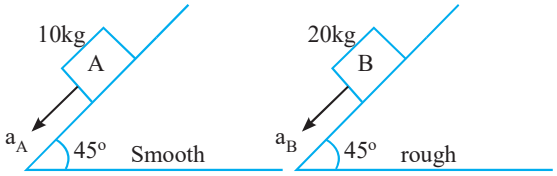
217. A block of mass 10 kg is kept on a fixed rough (μ = 0.8) inclined plane of angle of indication 30°. The frictional force acting on the block is



(a) 50 N (b) $50\sqrt{3}$N

(c) 52 N (d) 54 N

218. The ratio of the acceleration of blocks A placed on smooth incline with block B placed on rough incline is 2 : 1. The coefficient of kinetic friction between block B and incline is



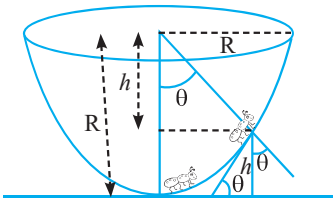
(a) 0.5 (b) 0.75

(c) 0.57 (d) None of these

219. A block of mass *m* is placed on a surface with a vertical cross-section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5 . The maximum height above the ground at which the block can be placed without slipping is

(a) $\frac{1}{6}$ m (b) $2\frac{2}{3}$ m (c) $\frac{1}{3}$ m (d) $\frac{1}{2}$ m

220. Find height from where Chiti will lose contact of the bowl and slip down. the coefficient of friction between Chiti and surface of bowl is μ.

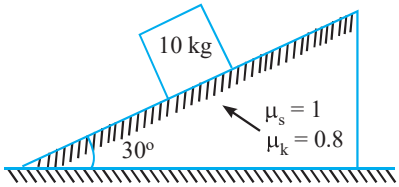


221. A block of mass m placed on an inclined plane of angle of inclination θ slides down the plane with constant speed. The coefficient of kinetic friction between block and inclined plane is

(a) sin θ (b) cos θ

(c) tan θ (d) $\tan^{-1}\theta$

222. A block of 10 kg mass is placed on a rough inclined surface as shown in figure. The acceleration of the block will be



(a) Zero (b) g (c) $\frac{g}{2}$ (d) $\frac{\sqrt{3}g}{2}$

223. The upper half of an inclined plane of inclination θ is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and lower half of the plane is given by

(a) $\mu = \frac{2}{\tan\theta}$ (b) μ = 2 tan θ

(c) μ = tan θ (d) $\mu = \frac{1}{\tan\theta}$

224. A block of 3 kg rests in limiting equilibrium on an inclined plane of inclination 30°. If the plane is raised to an inclination 60°, the force along the plane required to support it is at rest.

(a) $\frac{10}{\sqrt{3}}$ N (b) $10\sqrt{3}N$

(c) 10 N (d) $\frac{10}{3}$ N

225. A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 N, the mass of the block (in kg) is (take g = 10m /s^2)

(*a*) 2.0 (*b*) 4.0 (*c*) 1.6 (*d*) 2.5

226. The force required just to move a body up an inclined plane is double the force required just to prevent the body sliding down. If the coefficient of friction is 0.25, the angle of inclination of the plane is

(*a*) 30° (*b*) 45°

(*c*) $-\tan^{-1}\left(\frac{1}{4}\right)$ (*d*) $\tan^{-1}\left(\frac{3}{4}\right)$

227. A body of mass 10 kg is lying on a rough inclined plane of inclination 37° and μ = 1/2 the minimum force required to pull the body up the plane is

(*a*) 80 N (*b*) 100 N

(*c*) 120 N (*d*) 60 N

228. A body is sliding down an inclined plane $\left(\mu=\frac{1}{2}\right)$. If the normal reaction is twice that of the resultant downward force along the incline, the inclination of plane is

(*a*) 15° (*b*) 30°

(*c*) 45° (*d*) 60°

229. A block is released from the top of an inclined plane of inclination θ. Its velocity at the bottom of plane is *v*. If it slides down a rough inclined plane of the same inclination, its velocity at bottom is *v*/2. The coefficient of friction is

(*a*) $\frac{3}{4}\cot\theta$ (*b*) $\frac{1}{4}\cot\theta$ (*c*) $\frac{1}{4}\tan\theta$ (*d*) $\frac{3}{4}\tan\theta$

230. A given object takes *n* times as much time to slide down a 45° rough incline as it takes down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is given by

(*a*) $\left(1-\frac{1}{n^2}\right)$ (*b*) $\frac{1}{1-n^2}$ (*c*) $\sqrt{\left(1-\frac{1}{n^2}\right)}$ (*d*) $\sqrt{\frac{1}{1-n^2}}$

231. An ice cube is kept on an inclined plane of angle 30°. The coefficient of kinetic friction between the block and incline plane is the 1 /√3 What is the acceleration of the block?

(*a*) Zero (*b*) 2 m/s^2

(*c*) 1.5 m/s^2 (*d*) 5 m/s^2

232. A Rocket is fired vertically from the earth with an acceleration of 2*g*, where *g* is gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with horizontal, a point object of mass *m* is kept. The minimum coefficient of friction μ_{min} between the mass and inclined surface such that the mass does not move is

(*a*) tan θ (*b*) 2 tan θ

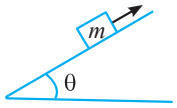
(*c*) 3 tan θ (*d*) tan 2 θ

233. A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. the magnitude of the contact force will be:

(*a*) Mg (*b*) Mg cos θ

(*c*) $\sqrt{Mg\sin\theta + Mg\cos\theta}$ (*d*) $Mg\sin\theta\sqrt{1+\mu}$

234. A block of mass m is thrown upwards with some initial velocity as shown. On the block



Column-I		Column-II	
1	Net force in horizontal direction	P	Zero
2	Net force in vertical direction	Q	m (g sin θ + μ cos θ)
3	Net force along the plane	R	m (g sinθcosθ + μg cos^{2}θ)
4	Net force perpendicular to plane	S	m (g sin^{2}θ + μg cosθ)

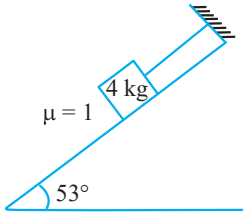
(*a*) (1) -P (2) -R (3) -Q (4)- P

(*b*) (1) -R (2) -S (3) -Q (4)- P

(*c*) (1) -P (2) -S (3) -Q (4)- P

(*d*) (1) -Q (2) -R (3) -P (4)- R

235. For the given figure, match column-I to column-II



Column-I		Column-II	
1	Tension in string	P	24 N
2	Normal on the block	Q	20 N
3	Contact force between block and inclined plane	R	12 N
4	Friction on the block	S	24 N

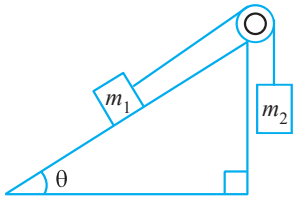
(*a*) (1) – Q (2) – P (3) – S (4) – R

(*b*) (1) – Q (2) – Q (3) – S (4) – R

(*c*) (1) – P (2) – Q (3) – R (4) – S

(*d*) (1) – Q (2) – P (3) – R (4) – S

236. Mass m_1 moves on a slope making an angle θ with the horizontal and is attached to mass m_2 by a string passing over a frictionless pulley as shown in figure. The coefficient of friction between m_1 and the slopping surface is μ. Which of the following statements are true?



(*a*) If $m_2 > m_1 \sin\theta$, the body will move up the plane

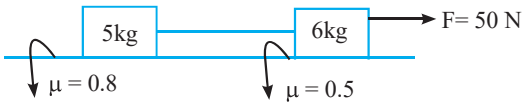
(*b*) If $m_2 > m_1 (\sin\theta + \mu\cos\theta)$, the body will move up the plane

(*c*) If $m_2 < m_1 (\sin\theta + \mu\cos\theta)$ the body will move up the plane

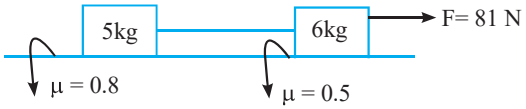
(*d*) If If $m_2 > m_1 (\sin\theta - \mu\cos\theta)$ the body will move down the plane

Connected Body Friction

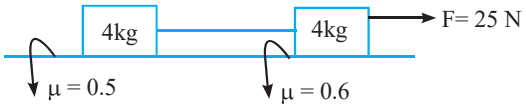
237. Two blocks of masses 5 kg and 6 kg are placed on a horizontal surface. The coefficient of friction between the 5 kg block and the surface is $\mu = 0.8$, and the coefficient of friction between the 6 kg block and the surface is $\mu = 0.5$. A force F = 50 N is applied horizontally to the 6 kg block. Find, the acceleration of the system. The tension in the string connecting the two blocks and the frictional forces acting on both the 5 kg and 6 kg blocks.



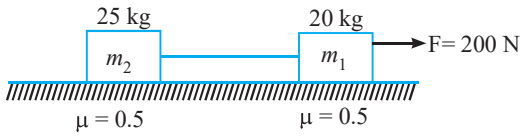
238. Find acceleration, tension in the string and friction acting on objects.



239. Find tension in the string, friction and acceleration.

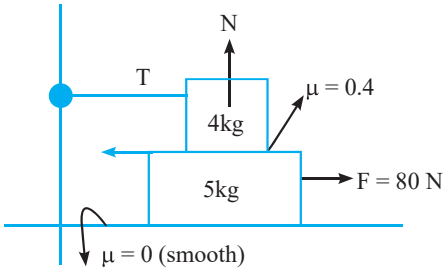


240. In the given figure, F = 200N, m_1= 20 kg m_2 = 25 kg, g = 10m /s^2 Find the friction acting on the block of mass m_2 [Take g = 10m /s ^{2}]



(*a*) 125 N (*b*) 120.5 N (*c*) 100 N (*d*) 123.2 N

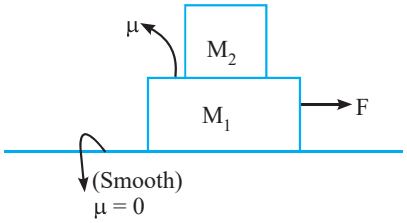
241. Find tension in string.



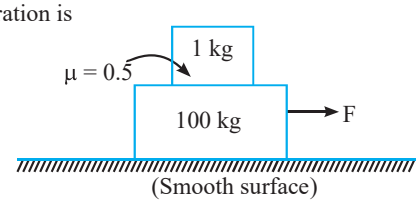
(*a*) T = 80N (*b*) T = 16N (*c*) T = 0 (*d*) T = 64 N

Friction in Block Over Block

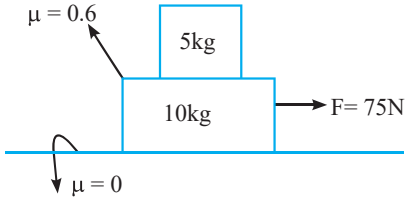
242. Two rough blocks system on smooth horizontal surface, find maximum force acting on lower block so that they will move together.



243. The maximum value of F which can be applied on the system shown in figure so that both blocks move together with same acceleration is



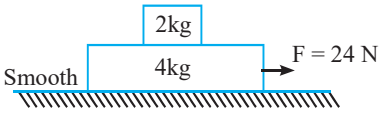
244. Find acceleration and friction acting on object. Will they move to gather



245. Two blocks A and B of masses 5 kg and 3 kg respectively rest on a smooth horizontal surface with B over A. The coefficient of friction between A and B is 0.5. The maximum horizontal force (in kg wt.) that can be applied to A, so that there will be motion of A and B without relative slipping is:

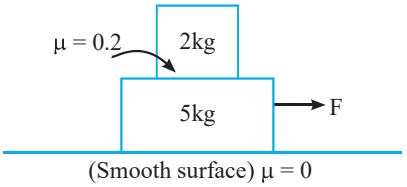
(*a*) 1.5 (*b*) 2.5 (*c*) 4 (*d*) 5

246. In the arrangement coefficient of friction between the two blocks $\mu = \frac{1}{2}$ is force of friction acting between the two blocks is:



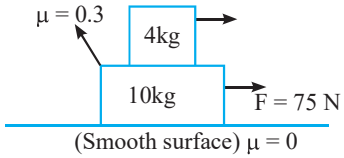
(*a*) 8 N (*b*) 6 N (*c*) 12 N (*d*) 10 N

247. For what value of force F, the mass 2 kg starts its motion in left direction relative to 5 kg [Take g = 10m/s^2]

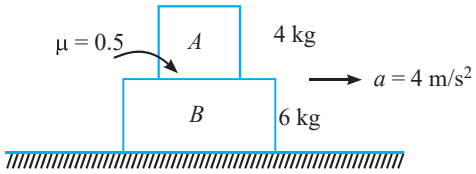


(*a*) 4 N (*b*) 10 N (*c*) 14 N (*d*) 7 N

248. A 4 kg block is placed on top of a 10 kg block. A horizontal force of 75 N is applied to the 10 kg block. The coefficient of friction between the two blocks is μ=0.3, while the surface beneath the 10 kg block is smooth. Find the acceleration of each block and the frictional force acting between the two blocks.

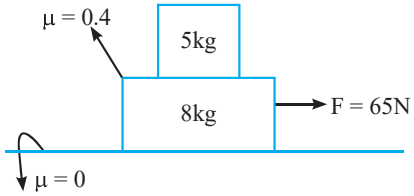


249. What is the friction force acting on the upper block when the system of two blocks moves with an acceleration of 4 m/s^2?

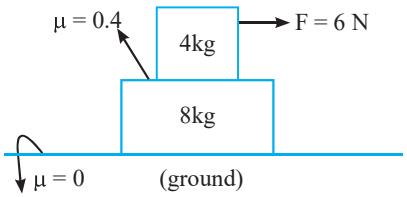


(*a*) 20 N (*b*) 16 N (*c*) 12 N (*d*) Zero

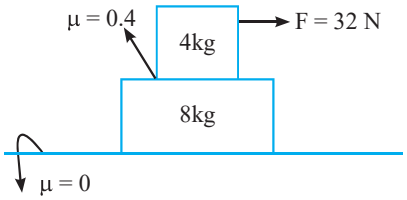
250. Find friction acting on 5 kg objects and acceleration of the system.



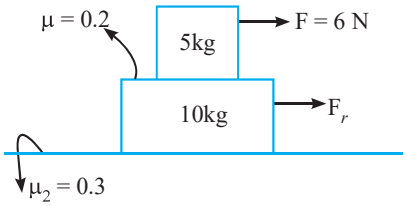
251. Find acceleration of the system and friction force acting on 4 kg object.



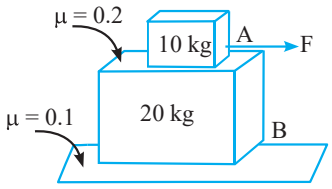
252. Find acceleration of both blocks and friction acting on objects



253. Find F so that they move togethet



254. In given diagram what is the minimum value of a horizontal external force Fon Block 'A' so that block 'B' will slide on ground is:



(*a*) 30 N (*b*) 20 N (*c*) 10 N (*d*) Not possible

Circular Dynamics

255. Object of mass' m is moving on a straight line path with constant speed then force on that object must be zero

(*a*) True (*b*) False

256. Object is moving on a circular path with constant speed then force on that object must be zero

(*a*) True (*b*) False

257. If a force of constant magnitude acts in direction perpendicular to the motion of a particle, then its

(*a*) Speed is uniform (*b*) Momentum is uniform

(*c*) Velocity is uniform (*d*) All of these

258. A vehicle of mass 200 kg is moving along a levelled curved road of radius 70 m with angular velocity of 0.2 rad/s. The centripetal force acting on the vehicle is:

(*a*) 560 N (*b*) 2800 N (*c*) 14 N (*d*) 2240 N

259. A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x}$ m/s^2 The value of x (Take $\pi = \frac{22}{7}$)

260. One end of string of length 1 is connected to particle of mass m and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed v, the net force on the particle (directed towards center) will be (T represents the tension in the string)

(*a*) Zero (*b*) T (*c*) $T + \frac{mv^2}{1}$ (*d*) $T - \frac{mv^2}{1}$

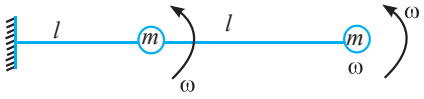
261. A stone of mass 1 kg is tied to end of a massless string of length 1 m. If the breaking tension of the string is 400 N, then maximum linear velocity, the stone can have without breaking the string, while rotation in horizontal plane, is:

(*a*) 20 ms^{-1} (*b*) 40 ms^{-1}

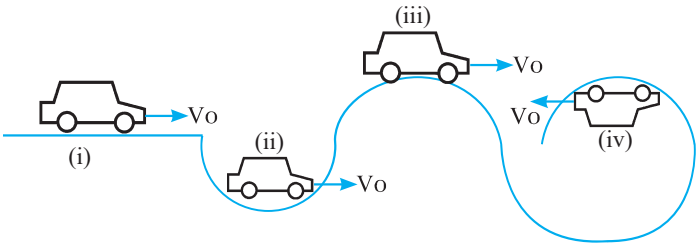
(*c*) 400 ms^{-1} (*d*) 10 ms^{-1}

262. Can centripetal and centrifugal act together?

263. Find tension T_1 & T_2 in both the string



264. Find normal reaction at each point

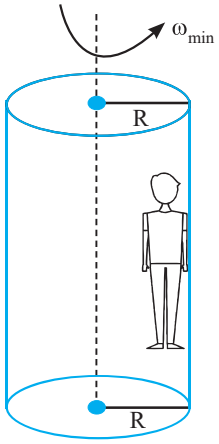


265. A roller coaster is designed such that riders experience "weightlessness" as they go round the top of a hill whose radius of curvature is 20 m. The speed of the car at the top of the hill is between

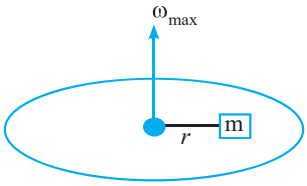
(*a*) 13 m/s and 14 m/s (*b*) 14 m/s and 15 m/s

(*c*) 15 m/s and 16 m/s (*d*) 16 m/s and 17 m/s

266. Find ω_{min} So that ramal of mass m in death wall of radius R, will not slip down.



267. An object of mass m is placed on a rotating disc at a distance r from the center. The coefficient of friction between the object and the disc is μ. Find the maximum angular velocity ω_{max} of the disc so that the object remains at rest with respect to the disc.



(a) $\sqrt{\frac{\mu g}{r}}$ (b) $\sqrt{\frac{g}{\mu r}}$ (c) $\frac{\mu g}{r}$ (d) $\frac{g}{\mu r}$

268. A boy is sitting on the horizontal platform of a joy wheel at a distance of 5 m from the center. The wheel begins to rotate and when the angular speed exceeds 1 rad/s, the boy just slips. The coefficient of friction between the boy and the wheel is ($g = 10 m/s^2$)

(a) 0.5 (b) 0.32 (c) 0.71 (d) 0.2

269. A gramophone record is revolving with an angular velocity ω. A coin is placed at a distance r from the centre of the record. The static coefficient of friction is μ. The coin will revolve with the record if

(a) $r = \mu g\omega^2$ (b) $r < \frac{\omega^2}{\mu g}$ (c) $r \le \frac{\mu g}{\omega^2}$ (d) $r \ge \frac{\mu g}{\omega^2}$

270. A coin placed on a rotating table just slips when it is placed at a distance of 1 cm from the center. If the angular velocity of the table in halved, it will just slip when placed at a distance of from the centre:

(a) 2 cm (b) 1cm (c) 8 cm (d) 4 cm

271. A particle of mass m is fixed to one end of a light spring of force constant k and unstretched length l. The system is rotated about the other end of the spring with an angular velocity ω, in gravity free space. The increase in length of the spring will be

(a) $\frac{m\omega^2 l}{k}$ (b) $\frac{m\omega^2 l}{k - m\omega^2}$

(c) $\frac{m\omega^2 l}{k + m\omega^2}$ (d) None of thes

272. A tube of length L. is filled completely with an incompressible liquid of mass M and closed at both end. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid at the other end is

(a) $\frac{1}{2}M\omega^2L^2$ (b) $m\omega^2L$ (c) $\frac{1}{4}M\omega^2L$ (d) $\frac{1}{2}M\omega^2L$

273. A train is moving with a speed of 12 m/s on rails which are 1.5 m apart. to negotiate a curve radius 400 m, the height by which the outer rail should be raised with respect to the inner rail is (Given, $g = 10 m/(s^2)$)

(a) 6.0 cm (b) 5. 4 cm (c) 4.8 cm (d) 4.2 cm

Bending of a Cyclist

274. A cyclist riding the bicycle at a speed of $14\sqrt{3}$ m/s takes a turn around a circular road of radius $20\sqrt{3}$ m without skidding. What is his inclination to the vertical?

(a) 30° (b) 45° (c) 60° (d) 75°

Motion of a Car on Rough horizontal Surface

275. A bus turns a slippery road having coefficient of friction of 0.5 with a speed of 10 m/s. The minimum radius of the arc in which bus turns is (Take $g = 10 m/s^2$)

(a) 4 m (b) 10 m (c) 15 m (d) 20 m

276. A car is moving on a horizontal circular track of radius 0.2 km with a constant speed. If coefficient of friction between tyres of car and road is 0.45, then speed of car may be (Take $g = 10 m/s^2$)

(a) 15 m/s (b) 30 m/s (c) 20 m/s (d) 40 m/s

277. A car is moving on a horizontal circular road of radius 0.1 km with constant speed. If coefficient of friction between tyres of car and road is 0.4, then speed of can not be ($g = 10 m/s^2$)

(a) 5 m/s (b) 10 m/s (c) 20 m/s (d) All of these

278. A car of mass m is moving on a level circular track of radius R. If μ_s represents the static friction between the road and tyres of the car, the maximum speed of the car in circular motion is given by

(a) $\sqrt{\mu_s mRg}$ (b) $\sqrt{\frac{Rg}{\mu_s}}$ (c) $\sqrt{\frac{mRg}{\mu_s}}$ (d) $\sqrt{\mu_s Rg}$

Motion of Car on Rough Banked Road

279. A car is negotiating a curved road of radius R. The road is banked at an angle θ, the coefficient of friction between the tyres of the car and the road is μ_s. The maximum safe velocity on this road is

(a) $\sqrt{gR^2\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$ (b) $\sqrt{gR\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

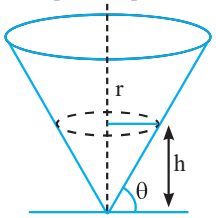
(c) $\sqrt{\frac{g}{r}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s + \tan\theta}\right)}$ (d) $\sqrt{\frac{g}{R^2}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

Conical Pendulum

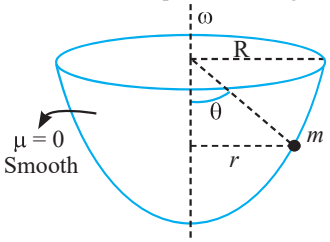
280. If the string of a conical pendulum makes an angle θ with horizontal, then square of its time period is proportional to

(a) $\sin\theta$ (b) $\cos\theta$ (c) $\tan\theta$ (d) $\cot\theta$

281. A particle describes a horizontal circle of radius r on the smooth surface of an inverted cone as shown. The height of plane of circle above vertex is h. The speed of particle should be



282. Find angle at which m is at equilibrium in given figure

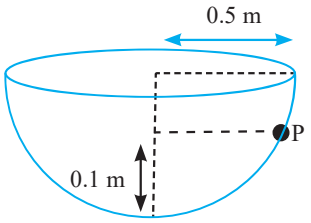


283. A hemispherical bowl of radius r is set rotating about its axis of symmetry in vertical. A small block kept in the bowl rotates with the bowl without slipping on its surface. If the surface of the bowl is smooth and the angle made by the radius through the block with the vertical is, θ then find the angular speed at which the ball is rotating.

(*a*) $\omega = \sqrt{rg\sin\theta}$ (*b*) $\omega = \sqrt{g / r\cos\theta}$

(*c*) $\omega = \sqrt{\frac{gr}{\cos\theta}}$ (*d*) $\omega = \sqrt{\frac{gr}{\tan\theta}}$

284. A small mass of 10 g lies in a hemispherical bowl of radius 0.5 m at a height of 0.1 m from the bottom of the bowl. The mass will be in equilibrium if the bowl rotates at an angle ω. Then find the value of ω.



285. A particle of mass m is suspended form a ceiling through a string of length *L*. The particle moves in a horizontal circle of radius *r* such that $r = \frac{L}{\sqrt{2}}$ The speed of particle will be

(*a*) $\sqrt{rg}$ (*b*) $\sqrt{rg/2}$ (*c*) $2\sqrt{rg}$ (*d*) $\sqrt{2rg}$

MR* CORNER

1. **Statement I:** If all Matter were made of electrically neutral particles such as Neutrons, It would be not possible to sit on a chair.

Statement II: If all Matter were made of electrically Neutral particles such as Neutrons, The earth could not move around the Sun.

Statement III: If all Matter were made of electric ally be neutral particles such as Neutrons. There would be ro tension in the string.

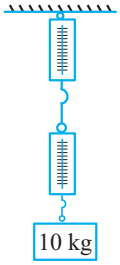
2. **Statement I:** Nuclear forces Between Two electrons at a given separation would be greater in magnitude than electromagnetic force between Two.

Statement II: When you stay on Trunk of tree, it exerts a frictional upward force on you to hold you there and you exert an equal frictional down word force on Tree.

Statement III: When a horse pulls a cart, the force that helps the horse to Move forward is the force exerted by the cort on the horse.

3. **Statement I:** The ceiling will pull the chain by $\frac{w_1 + w_2}{2}$ force if A Body of weight W_1 is suspended from the ceiling of a room through a chain of weight W_2.

Statement II: Both the scales will read 5 kg, if a Block of mass 10 kg is suspended through two light spring Balances as shown.



4. The force exerted by the smooth inclined wedge of inclination θ on the Block placed over it will reduced to _____Times when the whole system is accelerated horizontally so that the block do not slip on the wedge.

5. **Statement I:** A force F_1 acts on a particle so as to accelerate it from rest to velocity v. The force of F_2 which decelerates it to rest must be of equal magnitude as that of f_1.

Statement II: Two objects A and B (mass of A > B) are thrown upwards simultaneously with same speed. Then, A will go higher than B if the air exerts a constant and equal force of resistance on two Bodies.

6. A smooth wedge A is fitted in chamber hanging from a fixed ceiling near the earth's surface. A Block B placed at top of wedge takes Time T to slide down the length of wedge. of wedge. The Block will_____ when it is Released from Top of wedge & cable supporting the chamber is Broken at Same instant.

(*a*) Take longer Time than T to slide down the wedge.
(*b*) Remain at top of the wedge.
(*c*) Take shorter Time than T to slide down the wedge.
(*d*) Jump off the wedge.

7. **Statement I:** A person standing on floor of an elevator drops coin. The coin reaches the floor of elevator in a lesser time when lift is moving uniformly upwards as compared to time taken by the coin to reach the floor of stationary elevator.

Statement II: A person standing on floor of an elevator drops coin. The coin reaches the floor of elevator in a longer time when lift is moving uniformly downwards as compared to time taken by the coin to reach the floor of stationary elevator.

8. **Statement I:**A particle stays at Rest as seen in a frame. We cannot conclude that Resultant force on the particle must Be Zero.

Statement II: The frame May be non inertial but there is non zero resultant force.

9. The force exerted by the floor of an elevator on foot of a person standing there is more than the weight of the person if elevator is _____and_____.

(*a*) going up, slowing down
(*b*) going up, speeding up
(*c*) going down, slowing down
(*d*) going down, speeding up.
(*e*) Both (*b*) and (*c*)

10. **Statement I:** The Tension in the cable supporting an elevator must be greater than weight of elevator so that elevator goes upward with uniform speed.

Statement II: A person must be a liar when he says that he measured the acceleration of particle to be non zero while no force was acting on the particle.

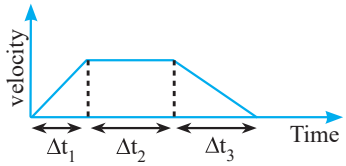
11. If the angle between the net contact Force having constant magnitude (provided by rough horizontal surface on a body placed on it) and vertical is decreased, the frictional force between the surface and body will_____.

12. **Statement I:** While walking on ice, one should take small steps to avoid slipping this is because smaller steps ensure larger amount of friction.

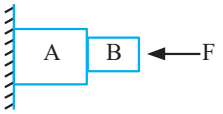
Statement II: Two cars of unequal masses use similar tyres. The minimum stopping distance will be same for both cases if they are moving at the same initial speed.

13. A Body of mass M is kept on α rough horizontal surface [coefficient of friction = μ]. person is trying to pull the body by applying horizontal force but body is not moving. The net force applied by surface on body is F, where maximum value of F is_____ and it's minimum value is_____.

14. A scooter Starting from rest moves according to the given graph. A 500 N man sitting on scooter behind driver manages to stay at rest with respect to scooter without touching any other part. The force exerted by seat on man is_____for time Δt_1,_____ for time Δt_2 and _____ for time Δt_3. (you have to compare w. r. t 500 N force, i.e. Greater than / equal to/ less than 500N)



15. **Statement I:** The following system cannot remain in equilibrium if wall is smooth but the surfaces of A and B in contact are rough.



Statement II: The direction of friction on B due to A will be in upward direction in order to achieve equilibrium considering all surfaces in above given figure to be rough.

16. **Statement I:** A block a kept on an inclined surface Just Begins to slide if inclination is 30° The block is replaced by another block B and it is found that it just begins to slide, if inclination is 40° from the following observation, it can be concluded that mass of A is greater than mass of B.

Statement II: The contact force exerted by a body A on another body B is equal to normal force between the bodies. We can conclude that the surfaces in contact must be frictionless.

17. **Statement I:** Static friction is always greater than the kinetic friction.

Statement II: limiting friction is never less than static & kinetic friction.

18. A Block is placed on rough floor and a horizontal force F is applied on it. The force of friction f by floor on block is measured for different values of F and a graph is plotted between them. The graph is a_____ for small F and is a_____ for large F.

19. **Statement I:** When a particle moves in circle with a uniform speed, it's velocity changes continuously as constant acceleration is Acting on it.

Statement II: The centrifugal force on a particle of mass m moving in a circle of radius r with a uniform speed v as observed from an inertial frame of reference will be $\frac{mv^2}{r}$ away from centre

20. A car moves at a constant speed on a road as shown in figure.



Consider the following statement I & II and state that whether They are True or False.

Statement I: The weight experienced by drives in car will be greater than its actual weight at point B and the observed weight will be lesser weight at point A and C.

Statement II: Normal force applied by the road on car at point C will be greater than force applied by it at point A.

21. The frame rotating about the Z- axis with a uniform angular speed ω_0 observes a particle of mass m rotating about the Z– axis in a circle of radius R with uniform angular speed ω. the centrifugal force on the particle will be.

22. **Statement I:** water in bucket is whirled in vertical circle with a string attached to it The water doesn't fall down even when the Bucket is inverted at Top of it's path. We can conclude that in This position, value of mg must be equal to $\frac{mv^2}{r}$.

Statement II: A Stone of mass m tied to a string of length l is rotated in a circle with the other end of the string at the centre. If string breaks, the stone will move away from centre due to centrifugal force.

23. A coin placed on a rotating Turntable just slips if it is placed at a distance of _____ from the centre. If the angular velocity of Turntable is doubled, it will just sleep at distance of 1 cm.

24. **Statement II:** Three identical cars A, B and C are moving at the samespeed on Three Bridges. The Car A goes on plane Bridge, B on Bridge convex upward and C on a bridge concave upwards. If F_A, F_B. & F_C are Normal forces exerted by cars on Bridges when they are at Middle of bridges, then F_C will be Maximum of Three forces.

Statement II: The position vector of a particle in a circular motion about the origin sweeps out equal area in equal time. It means the velocity of particle Remains constant.

25. A circular road of radius r is Banked for a speed = 40 km/hr. The friction coefficient Between the Tyre and road is negligible. Then

Statement I: If the car Turns at speed less than to 40 km/hr, it will slip down.

Statement II: If the Car turns at correct speed of 40 km/hr, the force applied by the Road on the car will be equal to $\frac{mv^2}{r}$

Answer Key

1. (b) **2.** (c) **3.** (c) **4.** (a) **5.** (a) **6.** (b) **7.** Due to inertia of motion **8.** (c)

9. (c) **10.** (a) **11.** Zero **12.** $\theta = \tan^{-1}\left(\frac{F}{mg}\right), T = \sqrt{(mg)^2 + f^2}$ **13.** (c) **14.** $x = 3$ **15.** (d) **16.** (b)

17. (d) **18.** (c) **19.** (b) **20.** (c) **21.** 130 N **22.** N = (mg –Fcosθ) **23.** 100 N **24.** 48 mg **25.** 10 N

26. 25 N **27.** (a) **28.** (a) **29.** 30 N **30.** 5 kg **31.** 40 kg **32.** (b) **33.** (a) **34.** (c) **35.** (b)

36. (a) **37.** (b) **38.** (d) **39.** (a) **40.** 8 kg **41.** $\frac{M_1}{M_2} = \frac{\sqrt{3}}{1}$ **42.** $f = \frac{mg\sin\theta}{1+\cos\theta}$ **43.** $x = 12$

44. (d) **45.** (b) **46.** (c) **47.** (b) **48.** $T = \frac{80}{3}N$ **49.** (c) **50.** $a_A = 2g$, $a_B = -g$ **51.** (a)

52. (c) **53.** (c) **54.** (c) **55.** (d) **56.** (a) **57.** (b) **58.** (b) **59.** (c) **60.** (b) **61.** (b)

62. (d) **63.** (c) **64.** f = 4 N **65.** $F_A : F_B = 1:3$ **66.** (d) **67.** (a) **68.** $4\sqrt{5}\ N.s$ **69.** (d)

70. (b) **71.** (a) **72.** (b) **73.** (b) **74.** (a) **75.** (a) **76.** (b) **77.** (c) **78.** (d) **79.** (a)

80. 6.67 kg/s **81.** (b) **82.** (c) **83.** (b) **84.** (a) **85.** (b) **86.** (c) **87.** (74kg m/s) **88.** 0.4 m/s

89. 2 bullets/sec **90.** (b) **91.** (c) **92.** 6000 **93.** (c) **94.** (c) **95.** (b) **96.** (c) **97.** 18 N

98. (d) **99.** 12 N, 13.5 N, 17.5 N **100.** (a) **101.** $T_1 = 64N$, $T_2 = 20N$, $a = 4m/s^2$ **102.** 42 N **103.** (c)

104. (b) **105.** $T_1 = 36$ N, $T_2 = 60$ N **106.** (c) **107.** (c) **108.** (b) **109.** (d) **110.** 10 N **111.** (b)

112. 240 N **113.** (b) **114.** (c) **115.** (c) **116.** $a = \frac{(m_2 - m_1)g}{m_1 + m_2}, T = \frac{2m_1 m_2}{m_1 + m_2}g$ **117.** 2 m_2g **118.** (d)

119. $a = \frac{(m_2 - m_1)g}{m_1 + m_2}$ **120.** $a = \frac{M_2 g}{M_1 + M_2}$ $T = \frac{M_1 M_2 g}{M_1 + M_2}$ **121.** (a) **122.** 3 m/s^2

123. (i) $\frac{M_2 g - M_1 g\sin\theta}{M_1 + M_2}$, (ii) $\frac{M_2 g\,\mathrm{Sin}\,\beta - M_1 g\sin\alpha}{M_1 + M_2}$, (iii) $\frac{10}{7} m/s^2$ **124.** (b) **125.** $x = 6$ **126.** (b) **127.** (c) **128.** (c)

129. 6 m/s^2 **130.** (b) **131.** $\frac{F + ma_0}{2m}$ **132.** (b) **133.** (b) **134.** (b) **135.** (i) $4a_2 = a_1$, (ii) $a_1 = -a_2$

136. $V_p = \frac{V}{4}$ **137.** (c) **138.** (c) **139.** (b) **140.** (d) **141.** (c) **142.** (c) **143.** (a) **144.** (d) **145.** (a)

146. 65 N w.r.t Both **147.** (a) **148.** (d) **149.** 12 N **150.** (d) **151.** (d) **152.** (d) **153.** (c) **154.** (c)

155. (d) **156.** (d) **157.** (b) **158.** (a) **159.** (d) **160.** (a) **161.** (a) **162.** (d) **163.** (d) **164.** (c)

165. (c) **166.** (b) **167.** (c) **168.** (b) **169.** (a) **170.** (a) **171.** (d) **172.** (a) **173.** $a = 0$ m/s^2 friction = 60 N

174. 5 N **175.** (d) **176.** $a = 2$m/s^2, friction = 20 N **177.** 16N **178.** $S = \frac{V_0^2}{2\mu g}$ **179.** (d) **180.** (b)

181. 10 N to 110N **182.** (F_{mr}) minimum = 40N, (F_{mr}) maximum =120 N **183.** (b) **184.** (d) **185.** (c)

186. (a) **187.** (c) **188.** (a) **189.** (a) **190.** (c) **191.** (d) **192.** (d) **193.** (b) **194.** (d) **195.** (a)

196. Friction = 8N, Tension = 16 N & Acceleration = 2m/s^2 **197.** 60 N, 60 N, 0 **198.** (c) **199.** (c)

200. $F = \frac{200}{7}N$ **201.** $F = \frac{mg}{\mu\sin\theta + \cos\theta}$ **202.** $F_{min} = 35.7$ **203.** (*a*) **204.** (*b*) **205.** (*c*) **206.** (*a*)

207. (*b*) **208.** (*b*) **209.** (*b*) **210.** (*b*) **211.** (*b*) **212.** (*a*) **213.** (*a*) **214.** $F_{min} = \mu mg\cos\theta - mg\sin\theta$

215. $F_{min} = mg\sin\theta - \mu mg\cos\theta$ **216.** (*a*) **217.** (*a*) **218.** (*a*) **219.** (*a*) **220.** $h = R\left[1 - \frac{1}{\sqrt{\mu^2+1}}\right]$ **221.** (*c*)

222. (*a*) **223.** (*b*) **224.** (*b*) **225.** (*a*) **226.** (*d*) **227.** (*b*) **228.** (*c*) **229.** (*d*) **230.** (*a*) **231.** (*a*)

232. (*a*) **233.** (*a*) **234.** (*b*) **235.** (*a*) **236.** (*b*) **237.** $a = 0$, $T = 20$ N, $f_{5kg} = 20$ N, $f_{6kg} = 30$ N

238. $a = 1 m/s^2$, $T = 45$ N, $f_{5kg} = 40$, $f_{6kg} = 30$ N **239.** $T = 1$N, $f_{4kg} = 1$ N, $f_{4kg} = 24$ N, $a = 0$ **240.** (*c*) **241.** (*b*)

242. $F_{max} = (m_1 + m_2)\mu g$ **243.** 505 N **244.** $a = 5 m/s^2$, $f_r = 25$N, they move together **245.** (*c*) **246.** (*a*) **247.** (*c*)

248. for 4 kg block, $a = 3 m/s^2$ for 10 kg block, $a = 6.3 m/s^2$ 12 N **249.** (*b*) **250.** 20 N, 5 m/s^2 **251.** 0.5 m/s^2, 4N

252. $a_8 = 2 m/s^2$, $a_4 = 4 m/s^2$, $f_r = 16$ N **253.** 75 N **254.** (*d*) **255.** (*b*) **256.** (*b*) **257.** (*a*) **258.** (*a*)

259. $x = 125$ **260.** (*b*) **261.** (*a*) **262.** No **263.** $T_1 = \omega^2(2l)$, $T_2 = 3\omega^2 l$

264. (*i*)N = mg, (*ii*) $N = mg + \frac{mV_0^2}{R}$, (*iii*) $N = mg - \frac{mV_0^2}{R}$, (*iv*) $N = \frac{mV_0^2}{R} - mg$ **265.** (*b*) **266.** $\sqrt{\frac{g}{\mu R}}$ **267.** (*a*) **268.** (*a*)

269. (*c*) **270.** (*d*) **271.** (*b*) **272.** (*d*) **273.** (*b*) **274.** (*c*) **275.** (*d*) **276.** (*b*) **277.** (*d*) **278.** (*d*)

279. (*b*) **280.** (*a*) **281.** $V = \sqrt{gh}$ **282.** $\theta = \tan^{-1}\left(\frac{\omega^2 r}{g}\right)$ **283.** (*b*) **284.** 5 *rad/s* **285.** (*a*)

MR* CORNER

1. I. - True, II. - False, III. - True **2.** I. - False, II. - True, III. - True **3.** Both are False

4. Initially when not accelerated. Ni = mg cos θ when accelerated by a = gtan θ, $N_2 = \frac{mg}{\text{Cos}\,\theta}$, $N_2 = \frac{N_1}{\text{Cos}^2\theta}$ **5.** I. - False, II. - True

6. (*b*) **7.** Both False **8.** Both are Correct and R is correct Explanation of A. **9.** (*e*) Both *b* and *c*

10. Both are false **11.** Decrease **12.** I. - False, II. - True **13.** $F_{max} = Mg\sqrt{\mu^2+1}$, $F_{min} = Mg$

14. Greater than 500N for time Δt_1, 500N for time Δt_2, Greater than 500N for time Δt_3 **15.** Both are True

16. Both are False **17.** I. False, II. True **18.** Straight line with slope 45°, straight line parallel to the *F*-axis

19. Both are False **20.** Both are True **21.** $m\omega_0^2 R$ **22.** Both are false **23.** 4cm **24.** I. = True, II. = False

25. I. = True, II. = False